Time-varying volatility spillovers and extreme risk contagion between stablecoin markets and stock markets: Evidence from TVP-VAR and CoVaR approaches

稳定币市场与股票市场之间的时变波动溢出与极端风险传染：来自 TVP-VAR 与 CoVaR 方法的证据

Yi You[a,*]

[a]*School of Finance, XXX University*

Abstract

This paper investigates the time-varying volatility spillovers and extreme risk contagion effects between stablecoin markets and U.S. stock markets from April 2021 to June 2026. Using the time-varying parameter vector autoregression (TVP-VAR) connectedness approach proposed by Antonakakis et al. [4], we first construct a dynamic spillover index framework that overcomes the rolling-window bias in traditional Diebold-Yilmaz spillover measures. We further decompose spillovers into frequency domains using the Baruník-Křehlík approach and quantify extreme tail risk contagion via the CoVaR and MES measures. Our empirical results reveal four key findings: First, stablecoin markets are materially connected to traditional financial markets, with a full-sample average total spillover index of 47.2%, indicating substantial connectedness once exchange and crypto-exposed equities are included in the system. Second, spillover effects exhibit pronounced time variation, ranging from 29.9% to 90.0%; the largest spike (87.5%) occurs during the March 2023 Silicon Valley Bank collapse and the associated USDC depegging, whereas crypto-native crises such as the Terra-LUNA collapse and the FTX bankruptcy generate only moderate increases (around 47–48%). Third, frequency decomposition shows that volatility spillovers are dominated by the low-frequency (long-term, 21+ days) component, which accounts for 48.6% of total spillovers, against only 20.0% at the short-term (1–5 days) band, indicating that persistent fundamental linkages dominate high-frequency transmission. Fourth, tail risk analysis demonstrates that extreme stablecoin shocks exert significant contagion effects on stock market downside risk, with crypto-concept stocks (ΔCoVaR $= -1.26\%$) and exchange stocks (-1.06%) exhibiting far larger tail exposures than payment stocks (-0.29%), the broad market (-0.23%), and bank stocks (statistically insignificant). The stablecoin block (USDT and USDC) is a net volatility transmitter to equities throughout the sample, and its net transmission strengthens further during crisis windows. Our findings have important implications for financial regulators monitoring systemic risk from digital asset markets and for portfolio managers constructing cross-asset hedging strategies.

*Corresponding author.

Keywords: Stablecoins, Volatility spillovers, TVP-VAR, CoVaR, Tail risk, Financial stability

1. Introduction / 引言

Since the launch of Tether (USDT) in 2014, stablecoins have experienced explosive growth, with total market capitalization rising from under $10 billion in early 2020 to over $250 billion by mid-2026. As digital tokens designed to maintain a stable value relative to fiat currencies (predominantly the U.S. dollar), stablecoins were initially envisioned as a "safe haven" within the volatile cryptocurrency ecosystem and a bridge between decentralized finance (DeFi) and traditional financial markets. However, the rapid expansion of stablecoin markets and their increasing integration with conventional finance have raised significant concerns among policymakers and financial regulators about potential systemic risk transmission channels.

自 2014 年泰达币（USDT）推出以来，稳定币经历了爆发式增长，其总市值从 2020 年初的不足 100 亿美元上升至 2026 年年中的 2500 亿美元以上。作为旨在相对法定货币（主要是美元）保持稳定价值的数字代币，稳定币最初被设想为动荡的加密货币生态系统中的"避风港"，以及连接去中心化金融（DeFi）与传统金融市场的桥梁。然而，稳定币市场的迅速扩张及其与传统金融日益加深的融合，引发了政策制定者和金融监管机构对潜在系统性风险传导渠道的重大关切。

The 2022 Terra-LUNA collapse, the 2022 FTX bankruptcy, and the March 2023 depegging of USD Coin (USDC) following the Silicon Valley Bank (SVB) failure vividly demonstrated that instabilities in the digital asset space can quickly propagate across markets. During these episodes, stablecoin redemption pressures led to fire sales of reserve assets (including U.S. Treasury bills and bank deposits), while sharp movements in stablecoin market capitalization and peg prices coincided with drawdowns in crypto-related equities and broader risk assets. Most recently, the passage of the U.S. GENIUS Act (Guiding and Establishing National Innovation for U.S. Stablecoins Act) in the Senate in June 2025 and its signing into law in July 2025 have further highlighted the interconnectedness between stablecoin regulation and traditional financial institutions. If banks and payment companies are permitted to issue their own stablecoins, the linkages between stablecoin markets and equity markets will only strengthen.

2022 年 Terra-LUNA 崩盘、2022 年 FTX 破产，以及 2023 年 3 月硅谷银行（SVB）倒闭后 USD Coin（USDC）的脱锚事件生动地表明，数字资产领域的不稳定可以迅速跨市场传导。在这些事件中，稳定币赎回压力导致储备资产（包括美国短期国债和银行存款）的抛售，而稳定币市值与锚定价格的剧烈波动与加密相关股票及更广泛风险资产的回撤同步出现。最近，美国《GENIUS 法案》(《指导和建立美国稳定币国家创新法案》）于 2025 年 6 月在参议院通过并于 2025 年 7 月签署成为法律，进一步凸显了稳定币监管与传统金融机构之间的相互联系。如果银行和支付公司获准发行自己的稳定币，稳定币市场与股票市场之间的联系只会进一步增强。

Despite growing policy attention, the existing literature on stablecoins remains primarily focused on their role within cryptocurrency markets [12], their reserve management practices [17], and their impact on foreign exchange markets [15]. Surprisingly few studies have systematically examined the volatility spillovers and extreme risk contagion between stablecoin markets and traditional stock markets. Several important questions remain unanswered: First, to what extent do shocks to stablecoin markets transmit to equity markets? Second, how do spillover effects vary over time, particularly during crisis events and regulatory announcements? Third, do spillovers operate through short-term information channels or long-term fundamental linkages? Fourth, which stock market sectors are most vulnerable to extreme stablecoin shocks?

尽管政策关注日益增加，现有关于稳定币的文献仍主要集中于其在加密货币市场中的作用 [12]、其储备管理实践 [17] 以及其对外汇市场的影响 [15]。很少有研究系统地考察稳定币市场与传统股票市场之间的波动溢出与极端风险传染。若干重要问题仍未得到回答：第一，稳定币市场的冲击在多大程度上会传导至股票市场？第二，溢出效应如何随时间变化，特别是在危机事件和监管公告期间？第三，溢出是通过短期信息渠道还是长期基本面联系发挥作用？第四，哪些股票市场板块最容易受到稳定币极端冲击的影响？

This paper fills these gaps in the literature by making the following contributions: First, to the best of our knowledge, this is the first comprehensive study to examine both average volatility spillovers and tail risk contagion between stablecoin markets and disaggregated stock market sectors (banks, exchanges, payment companies, and crypto-concept stocks). Second, we employ the TVP-VAR connectedness approach of Antonakakis et al. [4], which allows us to capture time-varying spillovers without the arbitrary window-size selection that plagues traditional rolling-window Diebold-Yilmaz estimation. Third, we complement the spillover analysis with frequency-domain decomposition using the method of Baruník and Křehlík [5], enabling us to distinguish between short-term (1–5 days), medium-term (6–20 days), and long-term (21+ days) spillover components. Fourth, we quantify extreme tail risk contagion using the CoVaR framework of Adrian and Brunnermeier [3] and the Marginal Expected Shortfall (MES) measure of Acharya et al. [2], providing evidence on how stablecoin stress events affect stock market downside risk. Finally, we conduct detailed heterogeneity analysis across sectors and market states, revealing rich cross-sectional variation in vulnerability to stablecoin shocks.

本文通过以下贡献填补文献空白：第一，据我们所知，这是首个系统考察稳定币市场与细分股票市场板块（银行、交易所、支付公司和加密货币概念股）之间平均波动溢出与尾部风险传染的综合研究。第二，我们采用 Antonakakis et al. [4] 的 TVP-VAR 关联性方法，能够在不依赖传统滚动窗口 Diebold-Yilmaz 估计所面临的任意窗口长度选择的情况下捕捉时变溢出。第三，我们使用 Baruník and Křehlík [5] 的方法对溢出进行频域分解，从而区分短期（1–5 天）、中期（6–20 天）和长期（21 天以上）溢出成分。第四，我们利用 Adrian and Brunnermeier [3] 的 CoVaR 框架和 Acharya et al. [2] 的边际期望损失（MES）测度量化极端尾部风险传染，为稳定币压力事件如何影响股票市场下行风险提供证据。最后，我们进行了跨板块和跨市场状态的详细异质性分析，揭示了各板块对稳定币冲击脆弱性的丰富截面差异。

Our empirical analysis, based on a daily sample from April 15, 2021 to June 30, 2026 (1,288 trading days), yields several important findings. First, we find substantial average connectedness between stablecoin markets and stock markets, with a total spillover index of 47.2%. The VIX and USDC are the largest net transmitters of volatility in the system, while the S&P 500 and crypto-concept stocks are the largest net receivers. Second, spillover effects are highly time-varying, with the index ranging between 29.9% and 90.0%. The single largest spike occurs around the March 2023 SVB failure and the associated USDC depegging, when the total spillover index reached 87.5% and remained above 80% for several months; by contrast, the crypto-native Terra-LUNA and FTX crises produced only moderate increases in economy-wide connectedness. Third, frequency decomposition reveals that nearly half (48.6%) of all spillovers are attributable to the long-term (21+ days) frequency band, while short-term (1–5 days) spillovers account for only 20.0%, indicating that persistent, slow-moving linkages dominate high-frequency sentiment transmission. Fourth, tail risk analysis shows that extreme stablecoin market declines are associated with significantly increased downside risk in crypto-exposed equity sectors: crypto-concept stocks exhibit a ΔCoVaR of -1.26% and an MES of -2.54%, several times larger in magnitude than those of payment stocks and the broad market, while bank stocks show no statistically significant tail exposure. Finally, the stablecoin block acts as a net volatility transmitter to equities throughout the sample, with net spillovers rising from 48.9 percentage points in tranquil periods to 57.3 percentage points during crisis windows.

我们的实证分析基于 2021 年 4 月 15 日至 2026 年 6 月 30 日的日度样本（1,288 个交易日），得出若干重要发现。第一，稳定币市场与股票市场之间存在显著的平均关联性，总溢出指数为 47.2%。VIX 和 USDC 是系统中最大的波动净输出者，而标准普尔 500 指数和加密货币概念股是最大的净接收者。第二，溢出效应高度时变，总溢出指数介于 29.9% 至 90.0% 之间。最大的一次飙升出现在 2023 年 3 月 SVB 倒闭及 USDC 脱锚事件前后，总溢出指数达到 87.5% 并在数月内保持在 80% 以上；相比之下，Terra-LUNA 和 FTX 等加密原生危机仅引起全系统关联性的温和上升。第三，频域分解显示，近一半（48.6%）的溢出可归因于长期（21 天以上）频段，而短期（1–5 天）溢出仅占 20.0%，表明持久的、缓慢变化的基本面联系主导了高频情绪传导。第四，尾部风险分析表明，稳定币市场的极端下跌与加密暴露股票板块下行风险的显著上升相关：加密货币概念股的 ΔCoVaR 为 -1.26%、MES 为 -2.54%，其绝对值数倍于支付股和大盘，而银行股的尾部暴露不显著。最后，稳定币板块在整个样本期内始终是相对股市的波动净输出者，其净溢出从平静时期的 48.9 个百分点上升至危机窗口的 57.3 个百分点。

The remainder of the paper is organized as follows: Section 2 reviews the related literature. Section 3 outlines the theoretical transmission mechanisms between stablecoin markets and stock markets. Section 4 presents the econometric methodology. Section 5 describes the data and provides preliminary analysis. Section 6 reports the main empirical results. Section 7 presents heterogeneity analysis. Section 8 discusses robustness checks. Section 9 concludes with policy implications.

本文其余部分安排如下：第 2 节回顾相关文献；第 3 节概述稳定币市场与股票市场之间的理论传导机制；第 4 节介绍计量经济学方法；第 5 节描述数据并进行初步分析；第 6 节报告主要实证结果；第 7 节进行异质性分析；第 8 节讨论稳健性检验；第 9 节总结全文并提出政策含义。

2. Literature review / 文献综述

Our paper relates to three strands of literature. First, we contribute to the rapidly growing literature on stablecoins and their role in financial markets. Early studies focused on whether stablecoins actually maintain their pegs [17] and their function as safe havens within crypto markets [19]. Griffin and Shams [12] find that Tether minting is used to manipulate Bitcoin prices, particularly during market downturns. Kristoufek et al. [15] examine spillovers from stablecoin inflows to foreign exchange markets, documenting significant effects on emerging market currencies. However, the existing literature has not systematically examined stablecoin spillovers to equity markets at the sectoral level.

本文与三支文献相关。第一，我们为关于稳定币及其在金融市场中作用的快速增长的文献做出贡献。早期研究关注稳定币是否真正维持其锚定 [17] 及其在加密市场内的避风港功能 [19]。Griffin and Shams [12] 发现泰达币的增发被用于操纵比特币价格，尤其在市场下行期间。Kristoufek et al. [15] 考察了稳定币流入对外汇市场的溢出，记录了对新兴市场货币的显著影响。然而，现有文献尚未在板块层面系统考察稳定币对股票市场的溢出。

Second, our paper builds on the extensive literature using network connectedness measures to study financial market spillovers. Building on the seminal work of Diebold and Yilmaz [9, 10], numerous studies have examined volatility spillovers across asset classes [8], international stock markets [20], and financial institutions [11]. The recent development of time-varying parameter VAR approaches by Antonakakis et al. [4] has addressed important limitations of rolling-window estimation, while Baruník and Křehlík [5] extended the framework to the frequency domain. Our paper applies these methods to the novel context of stablecoin-equity market linkages.

第二，本文建立在利用网络关联性测度研究金融市场溢出的丰富文献之上。在 Diebold and Yilmaz [9, 10] 的开创性工作基础上，大量研究考察了跨资产类别 [8]、国际股票市场 [20] 以及金融机构之间 [11] 的波动溢出。Antonakakis et al. [4] 最近发展的时变参数 VAR 方法解决了滚动窗口估计的重要局限，而 Baruník and Křehlík [5] 将该框架扩展至频域。本文将这些方法应用于稳定币与股票市场关联这一全新情境。

Third, we contribute to the literature on tail risk and systemic risk measurement. Since the global financial crisis, a large literature has developed measures of systemic risk including CoVaR [3], MES [2], and SRISK [7]. These measures have been applied to study contagion across banks [16], between banks and sovereigns [1], and across cryptocurrencies [6]. We extend this literature by quantifying tail risk contagion from stablecoin markets to different stock market sectors, providing the first evidence on how stablecoin stress events affect equity market downside risk at the sectoral level.

第三，我们为尾部风险与系统性风险测度的文献做出贡献。自全球金融危机以来，大量文献发展了系统性风险测度方法，包括 CoVaR [3]、MES [2] 和 SRISK [7]。这些测度已被应用于研究银行之间 [16]、银行与主权之间 [1] 以及加密货币之间 [6] 的传染。我们通过量化稳定币市场对不同股票市场板块的尾部风险传染来拓展这支文献，首次在板块层面提供稳定币压力事件如何影响股票市场下行风险的证据。

3. Theoretical transmission mechanisms / 理论传导机制

Before proceeding to the empirical analysis, we outline three key theoretical channels through which stablecoin market shocks can transmit to stock markets.

在进行实证分析之前，我们概述稳定币市场冲击传导至股票市场的三条关键理论渠道。

3.1. Liquidity flow channel / 流动性流动渠道

The first and most direct channel operates through cross-market liquidity flows. Investors who wish to gain exposure to cryptocurrencies typically convert fiat currency into stablecoins on centralized exchanges, which then serve as “on-ramps” to the broader crypto ecosystem. Conversely, when investors reduce their crypto exposure, they convert back to stablecoins before redeeming for fiat currency. Importantly, a growing share of stablecoin reserve assets consists of traditional financial assets including U.S. Treasury bills, commercial paper, and bank deposits. When stablecoin issuers mint or redeem coins, they must buy or sell these reserve assets, which can directly affect prices in fixed income and credit markets, with knock-on effects to equity markets. For payment companies and banks that hold stablecoin reserves or offer stablecoin-related services, their equity valuations are directly exposed to stablecoin transaction volumes and market capitalization.

第一条也是最直接的渠道通过跨市场流动性流动发挥作用。希望获得加密货币敞口的投资者通常在中心化交易所将法定货币兑换为稳定币，稳定币随后成为通往更广泛加密生态系统的“入口”。反之，当投资者减少加密敞口时，他们先兑换回稳定币，再赎回为法定货币。重要的是，稳定币储备资产中越来越多的部分由传统金融资产构成，包括美国短期国债、商业票据和银行存款。当稳定币发行方铸造或赎回代币时，他们必须买入或卖出这些储备资产，这会直接影响固定收益和信用市场的价格，并对股票市场产生连锁影响。对于持有稳定币储备或提供稳定币相关服务的支付公司和银行而言，其股票估值直接暴露于稳定币交易量与市值的变化。

3.2. Sentiment and risk preference channel / 情绪与风险偏好渠道

The second channel operates through investor sentiment and risk preferences. Stablecoin market capitalization and on-chain activity are widely followed barometers of crypto market conditions. Increases in stablecoin issuance can signal either (i) new capital entering the crypto ecosystem (“risk-on” sentiment), which may boost crypto-concept stocks such as Coinbase, MicroStrategy, and Bitcoin miners, or (ii) investors moving to “cash” within crypto (“risk-off” sentiment), which may be associated with declines in risky assets more broadly. During extreme stress events such as stablecoin depeggings, investor risk aversion can spike, leading to selloffs not just in crypto assets but across risky asset classes including equities. This sentiment channel is likely to be particularly strong for stocks with high retail ownership or perceived crypto exposure.

第二条渠道通过投资者情绪与风险偏好发挥作用。稳定币市值与链上活动是被广泛关注的加密市场状况晴雨表。稳定币发行量的上升可能意味着：(i) 新资金进入加密生态系统（“追逐风险”情绪），这可能提振 Coinbase、MicroStrategy 和比特币矿企等加密货币概念股；或者 (ii) 投资者在加密生态内部转向“现金”（“规避风险”情绪），这可能与更广泛风险资产的下跌相联系。在稳定币脱锚等极端压力事件期间，投资者风险厌恶可能急剧上升，导致不仅在加密资产中，而且在包括股票在内的各风险资产类别中出现抛售。对于散户持股比例高或被认为具有加密敞口的股票，这一情绪渠道可能尤为强劲。

3.3. Regulatory risk channel / 监管风险渠道

The third channel operates through regulatory expectations. Major stablecoin regulatory developments (such as the GENIUS Act debates in 2025) have differential implications for different sectors of the stock market. Clear regulatory guidelines that allow banks and payment companies to issue stablecoins would represent a positive revenue opportunity for these traditional financial institutions. In contrast, stricter securities regulation of crypto assets could negatively affect crypto exchanges and pure-play crypto-concept stocks. Regulatory actions targeting specific stablecoin issuers also convey information about the broader regulatory stance toward digital assets, which can be priced into all crypto-exposed equities.

第三条渠道通过监管预期发挥作用。重大稳定币监管进展（如 2025 年《GENIUS 法案》的立法辩论）对股票市场不同板块具有差异化影响。允许银行和支付公司发行稳定币的明确监管指引将为这些传统金融机构带来正面的收入机会。相反，对加密资产更严格的证券

监管可能对加密交易所和纯加密货币概念股产生负面影响。针对特定稳定币发行方的监管行动也传达了关于数字资产整体监管立场的信息，这些信息会被定价到所有具有加密敞口的股票中。

These three channels lead us to the following testable hypotheses:

这三条渠道引导我们提出以下可检验假说：

H1 (Connectedness hypothesis): There are significant volatility spillovers between stablecoin markets and stock markets, with the strength of connectedness time-varying and intensifying during crisis events.

H1（关联性假说）：稳定币市场与股票市场之间存在显著的波动溢出，且关联强度随时间变化并在危机事件期间加剧。

H2 (Heterogeneity hypothesis): Spillover effects and tail risk exposure differ significantly across stock market sectors, with crypto-concept stocks exhibiting the highest sensitivity to stablecoin shocks, followed by exchange stocks, payment stocks, and traditional banks.

H2（异质性假说）：溢出效应与尾部风险暴露在股票市场各板块之间存在显著差异，其中加密货币概念股对稳定币冲击最为敏感，其次是交易所股票、支付股票和传统银行。

H3 (Frequency hypothesis): Cross-market spillovers are dominated by short-term (high-frequency) components rather than long-term components, consistent with sentiment and trading-driven transmission rather than fundamental economic linkages.

H3（频率假说）：跨市场溢出由短期（高频）成分而非长期成分主导，与情绪和交易驱动的传导机制相一致，而非基本面经济联系。

H4 (Tail risk hypothesis): Extreme negative stablecoin shocks exert disproportionately large contagion effects on stock market downside risk, particularly for crypto-exposed sectors.

H4（尾部风险假说）：稳定币的极端负向冲击对股票市场下行风险产生不成比例的传染效应，尤其是对具有加密敞口的板块。

4. Methodology / 研究方法

4.1. TVP-VAR connectedness approach / TVP-VAR 关联性方法

We begin by measuring volatility spillovers using the TVP-VAR connectedness framework developed by Antonakakis et al. [4], which extends the Diebold-Yilmaz approach to allow for time-varying parameters without relying on rolling windows. The TVP-VAR(p) model is specified as:

我们首先使用 Antonakakis et al. [4] 提出的 TVP-VAR 关联性框架来度量波动溢出，该框架将 Diebold-Yilmaz 方法扩展为允许参数随时间变化，且不依赖滚动窗口。TVP-VAR(p) 模型设定为：

$$\mathbf{y}_t = \mathbf{B}_t \mathbf{z}_{t-1} + \mathbf{u}_t, \quad \mathbf{u}_t \sim N(\mathbf{0},\ _t) \tag{1}$$

where $\mathbf{y}_t$ is an $N \times 1$ vector of endogenous variables (volatilities), $\mathbf{B}_t$ is an $N \times Np$ matrix of time-varying coefficients, $\mathbf{z}_{t-1} = (\mathbf{y}'_{t-1}, \mathbf{y}'_{t-2}, \ldots, \mathbf{y}'_{t-p})'$ is an $Np \times 1$ vector of lagged variables, and $\mathbf{u}_t$ is an $N \times 1$ vector of heteroskedastic disturbances with time-varying covariance matrix $_t$.

其中，$\mathbf{y}_t$ 是 $N \times 1$ 维内生变量（波动率）向量，$\mathbf{B}_t$ 是 $N \times Np$ 维时变系数矩阵，$\mathbf{z}_{t-1} = (\mathbf{y}'_{t-1}, \mathbf{y}'_{t-2}, \ldots, \mathbf{y}'_{t-p})'$ 是 $Np \times 1$ 维滞后变量向量，$\mathbf{u}_t$ 是具有时变协方差矩阵 $\,_t$ 的 $N \times 1$ 维异方差扰动向量。

The time-varying parameters and covariance matrices evolve according to random-walk state equations, and the model is estimated using the Kalman filter with exponentially weighted forgetting factors κ_1 and κ_2 governing the rate of parameter and covariance drift, respectively. Following the recommendations of Antonakakis et al. [4] for daily financial data, we set $\kappa_1 = \kappa_2 = 0.99$, implying that the effective estimation window places exponentially declining weight on roughly the past 100 observations.

时变参数与协方差矩阵按照随机游走状态方程演化，模型使用带有指数加权遗忘因子 κ_1 和 κ_2 的卡尔曼滤波进行估计，二者分别控制参数与协方差漂移的速率。遵循 Antonakakis et al. [4] 对日度金融数据的建议，我们设定 $\kappa_1 = \kappa_2 = 0.99$，这意味着有效估计窗口对大约过去 100 个观测施加指数递减的权重。

After estimating the TVP-VAR, we compute the generalized forecast error variance decompositions (GFEVD) following Koop et al. [14] and Pesaran and Shin [18], which are invariant to variable ordering. The H-step-ahead GFEVD from variable j to variable i is given by:

估计 TVP-VAR 之后，我们按照 Koop et al. [14] 和 Pesaran and Shin [18] 的方法计算广义预测误差方差分解（GFEVD），该方法对变量排序不变。由变量 j 到变量 i 的 H 步前向 GFEVD 为：

$$\phi^g_{ij,t}(H) = \frac{\sigma^{-1}_{jj,t} \sum_{h=0}^{H-1} (\mathbf{e}'_i \mathbf{A}_{h,t} \,_t \mathbf{e}_j)^2}{\sum_{h=0}^{H-1} (\mathbf{e}'_i \mathbf{A}_{h,t} \,_t \mathbf{A}'_{h,t} \mathbf{e}_i)} \tag{2}$$

where $\sigma_{jj,t}$ is the j-th diagonal element of $\,_t$, $\mathbf{e}_i$ is a selection vector with one in the i-th position, and $\mathbf{A}_{h,t}$ are the time-varying impulse response coefficient matrices at horizon h.
其中，$\sigma_{jj,t}$ 是 $\,_t$ 的第 j 个对角元素，$\mathbf{e}_i$ 是第 i 个位置为 1 的选择向量，$\mathbf{A}_{h,t}$ 是第 h 期的时变脉冲响应系数矩阵。

We normalize the GFEVD such that each row sums to one: $\tilde{\phi}^g_{ij,t}(H) = \phi^g_{ij,t}(H) / \sum_{j=1}^{N} \phi^g_{ij,t}(H)$. The total connectedness index (TCI), which measures the proportion of forecast error variance explained by cross-variable spillovers, is then computed as:

我们对 GFEVD 进行行归一化，使每行之和为 1：$\tilde{\phi}^g_{ij,t}(H) = \phi^g_{ij,t}(H) / \sum_{j=1}^{N} \phi^g_{ij,t}(H)$。衡量跨变量溢出占预测误差方差比例的总关联性指数（TCI）计算为：

$$\mathrm{TCI}_t(H) = \frac{\sum_{i=1}^{N} \sum_{j=1, j\neq i}^{N} \tilde{\phi}^g_{ij,t}(H)}{\sum_{i=1}^{N} \sum_{j=1}^{N} \tilde{\phi}^g_{ij,t}(H)} \times 100 = \frac{\sum_{i=1}^{N} \sum_{j=1, j\neq i}^{N} \tilde{\phi}^g_{ij,t}(H)}{N} \times 100 \tag{3}$$

We also compute directional spillovers FROM others TO variable i, $S_{i\leftarrow\cdot,t}(H) = \sum_{j\neq i} \tilde{\phi}^g_{ij,t}(H) \times 100$; directional spillovers TO others FROM variable j, $S_{\cdot\leftarrow j,t}(H) = \sum_{i\neq j} \tilde{\phi}^g_{ij,t}(H) \times 100$; and net spillovers for each variable, $S_{i,t}(H) = S_{\cdot\leftarrow i,t}(H) - S_{i\leftarrow\cdot,t}(H)$. A positive net spillover value indicates that variable i is a net transmitter of shocks to the system, while a negative value indicates it is a net receiver.

我们还计算：来自其他变量对变量 i 的方向性溢出 $S_{i\leftarrow\cdot,t}(H) = \sum_{j\neq i} \tilde{\phi}^g_{ij,t}(H) \times 100$；变量 j 对其他变量的方向性溢出 $S_{\cdot\leftarrow j,t}(H) = \sum_{i\neq j} \tilde{\phi}^g_{ij,t}(H) \times 100$；以及每个变量的净溢出

$S_{i,t}(H) = S_{\cdot\leftarrow i,t}(H) - S_{i\leftarrow\cdot,t}(H)$。净溢出为正表明变量 i 是系统中冲击的净输出者，为负则表明它是净接收者。

4.2. Frequency-domain connectedness / 频域关联性

To decompose spillovers by frequency, we follow Baruník and Křehlík [5] and define the frequency response function $(e^{-i\omega}) = \sum_h e^{-i\omega h}\mathbf{A}_h$ at frequency ω. The generalized spectral decomposition of the forecast error variance at frequency band $d = (a, b)$ is:

为按频率分解溢出，我们遵循 Baruník and Křehlík [5] 的方法，定义频率 ω 处的频率响应函数 $(e^{-i\omega}) = \sum_h e^{-i\omega h}\mathbf{A}_h$。在频段 $d = (a, b)$ 上预测误差方差的广义谱分解为：

$$(\boldsymbol{\Theta}_d)_{i,j} = \frac{1}{2\pi}\int_a^b \Gamma_i(\omega)(\mathbf{f}(\omega))_{i,j}d\omega \tag{4}$$

where $\mathbf{f}(\omega) = (e^{-i\omega})\ '(e^{+i\omega})$ is the spectral density of $\mathbf{y}_t$ at frequency ω, and $\Gamma_i(\omega)$ is the weighting function that scales the contribution by the share of variable i's variance explained at each frequency.

其中，$\mathbf{f}(\omega) = (e^{-i\omega})\ '(e^{+i\omega})$ 是 $\mathbf{y}_t$ 在频率 ω 处的谱密度，$\Gamma_i(\omega)$ 是权重函数，用于按各频率上解释的变量 i 方差份额对贡献进行缩放。

We can then compute frequency-specific total, directional, and net spillovers using the same formulas as above but with the frequency-band-specific variance decompositions $(\boldsymbol{\Theta}_d)_{i,j}$. We follow standard practice and define three frequency bands: short-term (1 to 5 days, corresponding to roughly one trading week), medium-term (6 to 20 days, one week to one month), and long-term (21 days and beyond, horizons longer than one month).

随后，我们使用与上文相同的公式、但以频段特定的方差分解 $(\boldsymbol{\Theta}_d)_{i,j}$ 计算特定频率的总溢出、方向性溢出与净溢出。遵循标准做法，我们定义三个频段：短期（1 至 5 天，约一个交易周）、中期（6 至 20 天，一周至一个月）和长期（21 天及以上，超过一个月的期限）。

4.3. CoVaR and MES for tail risk measurement / 尾部风险测度：*CoVaR* 与 *MES*

To measure extreme risk contagion from stablecoin markets to stock markets, we employ the Conditional Value-at-Risk (CoVaR) methodology of Adrian and Brunnermeier [3]. For a given confidence level q (we use $q = 5\%$), $\text{CoVaR}_q^{j|C(\text{SC})}$ is defined as the Value-at-Risk of institution (or portfolio) j conditional on stablecoin markets being in a crisis state $C(\text{SC})$:

为度量稳定币市场对股票市场的极端风险传染，我们采用 Adrian and Brunnermeier [3] 的条件在险价值（CoVaR）方法。对于给定的置信水平 q（我们取 $q = 5\%$），$\text{CoVaR}_q^{j|C(\text{SC})}$ 定义为在稳定币市场处于危机状态 $C(\text{SC})$ 的条件下机构（或组合）j 的在险价值：

$$\Pr(r_{j,t} \leq \text{CoVaR}_q^{j|C(\text{SC})} \mid C(\text{SC})) = q \tag{5}$$

We define the crisis state $C(\text{SC})$ as stablecoin basket returns being at their q-th quantile (i.e., large negative depegging returns). The marginal contribution of stablecoin distress to institution j's tail risk is given by ΔCoVaR:

我们将危机状态 $C(\text{SC})$ 定义为稳定币篮子收益处于其 q 分位数（即大幅负向脱锚收益）。稳定币危机对机构 j 尾部风险的边际贡献由 ΔCoVaR 给出：

$$\Delta\text{CoVaR}_q^{j|SC} = \text{CoVaR}_q^{j|r_{SC}=VaR_q^{SC}} - \text{CoVaR}_q^{j|r_{SC}=VaR_{50}^{SC}} \tag{6}$$

where VaR^{SC}_{50} is the median (normal state) of stablecoin returns. ΔCoVaR thus measures the additional tail risk of sector j when stablecoin markets are under stress relative to normal times.
其中，VaR^{SC}_{50} 是稳定币收益的中位数（正常状态）。因此，ΔCoVaR 衡量稳定币市场承压时板块 j 相对正常时期的额外尾部风险。

We estimate CoVaR using quantile regressions [13]:

我们使用分位数回归 [13] 估计 CoVaR：

$$r_{j,t} = \alpha_q + \beta_q \cdot r_{SC,t} + \gamma_q' \mathbf{M}_{t-1} + \varepsilon_t \tag{7}$$

where $\mathbf{M}_{t-1}$ is a vector of lagged state variables including the change in VIX, S&P 500 returns, the term spread (10-year minus 3-month Treasury), the credit spread (Baa corporate minus 10-year Treasury), and the 3-month Treasury bill rate. Then $\mathrm{VaR}^j_q(t) = \hat{\alpha}_q + \hat{\beta}_q \cdot VaR^{SC}_q(t) + \hat{\gamma}_q' \mathbf{M}_{t-1}$, and $\mathrm{CoVaR}^{j|SC}_q(t)$ is obtained by evaluating this expression at $r_{SC,t} = VaR^{SC}_q(t)$. Standard errors are obtained from 200 block bootstrap replications.
其中，$\mathbf{M}_{t-1}$ 是滞后状态变量向量，包括 VIX 变动、标准普尔 500 指数收益、期限利差（10 年期减 3 个月国债）、信用利差（Baa 公司债减 10 年期国债）以及 3 个月期国库券利率。于是 $\mathrm{VaR}^j_q(t) = \hat{\alpha}_q + \hat{\beta}_q \cdot VaR^{SC}_q(t) + \hat{\gamma}_q' \mathbf{M}_{t-1}$，将 $r_{SC,t} = VaR^{SC}_q(t)$ 代入该式即得 $\mathrm{CoVaR}^{j|SC}_q(t)$。标准误通过 200 次区组自助法（block bootstrap）获得。

As an alternative tail risk measure, we also compute the Marginal Expected Shortfall (MES) following Acharya et al. [2]:

作为另一种尾部风险测度，我们还按照 Acharya et al. [2] 计算边际期望损失（MES）：

$$\mathrm{MES}^j_q = -\mathbb{E}\left[r_{j,t} \mid r_{SC,t} \leq -VaR^{SC}_q\right] \tag{8}$$

MES measures the expected loss on sector j in the worst $q\%$ of stablecoin market outcomes. Both measures capture tail dependence, but CoVaR focuses on quantile co-movement while MES focuses on expected losses during extreme events.

MES 衡量在稳定币市场最差的 $q\%$ 情形下板块 j 的预期损失。两种测度都捕捉尾部相依性，但 CoVaR 侧重于分位数联动，而 MES 侧重于极端事件期间的预期损失。

5. Data and preliminary analysis / 数据与初步分析

5.1. Data sources and variable construction / 数据来源与变量构造

Our sample spans from April 15, 2021 to June 30, 2026, covering 1,288 trading days. The start date is determined by the listing of Coinbase (COIN) on April 14, 2021, which is required for the construction of the exchange sector portfolio; the sample therefore covers the 2021 crypto bull market, the 2022 bear market and Terra-LUNA collapse, the 2022 FTX bankruptcy, the March 2023 SVB failure and USDC depegging, the 2024–2025 crypto rally, and the 2025 GENIUS Act legislative process. We obtain daily data from multiple sources:

我们的样本期为 2021 年 4 月 15 日至 2026 年 6 月 30 日，共 1,288 个交易日。样本起点由 Coinbase（COIN）于 2021 年 4 月 14 日上市决定，该股票是构造交易所板块组合所必需的；因此样本覆盖了 2021 年加密牛市、2022 年熊市与 Terra-LUNA 崩盘、2022 年 FTX 破产、2023 年 3 月 SVB 倒闭与 USDC 脱锚、2024–2025 年加密市场反弹以及 2025 年《GENIUS 法案》立法进程。我们从多个来源获取日度数据：

Stablecoin market variables:. We collect daily market capitalization data for the largest stablecoins—Tether (USDT), USD Coin (USDC), Dai (DAI), and Binance USD (BUSD, until its phase-out)—from CoinGecko. We compute daily log changes in total stablecoin market capitalization (Δln(Stablecoin MC)) as our core quantity variable. In addition, we obtain daily closing prices of USDT/USD, USDC/USD, and DAI/USD from the Kraken exchange, and construct a supply-weighted stablecoin basket price return that captures de-pegging pressure; this basket return serves as the stablecoin market return $r_{SC,t}$ in the tail risk analysis. Conditional volatilities of USDT and USDC are estimated from univariate GARCH(1,1) models fitted to their daily price returns.

稳定币市场变量：. 我们从 CoinGecko 收集最大稳定币——泰达币（USDT）、USD Coin（USDC）、Dai（DAI）和币安 USD（BUSD，直至其退出）——的日度市值数据，并计算稳定币总市值的日度对数变化（Δln(Stablecoin MC)）作为核心数量变量。此外，我们从 Kraken 交易所获取 USDT/USD、USDC/USD 和 DAI/USD 的日收盘价，构造反映脱锚压力的供给加权稳定币篮子价格收益；该篮子收益在尾部风险分析中作为稳定币市场收益 $r_{SC,t}$。USDT 和 USDC 的条件波动率通过对其日度价格收益拟合单变量 GARCH(1,1) 模型估计得到。

Stock market variables:. We construct four sector portfolios from U.S. equities (equal-weighted, daily log returns): (i) a **bank portfolio** comprising JPMorgan Chase (JPM), Bank of America (BAC), Wells Fargo (WFC), Citigroup (C), Goldman Sachs (GS), and Morgan Stanley (MS); (ii) an **exchange portfolio** comprising Coinbase (COIN) and Robinhood (HOOD); (iii) a **payment portfolio** comprising PayPal (PYPL), Block (XYZ, formerly Square), Visa (V), and Mastercard (MA); and (iv) a **crypto-concept portfolio** comprising MicroStrategy (MSTR), Riot Platforms (RIOT), Marathon Digital (MARA), Hut 8 (HUT), and CleanSpark (CLSK). All stock data are obtained from Yahoo Finance. We also include the S&P 500 index (SPX) as the broad market benchmark and the CBOE VIX index as a measure of market uncertainty. Conditional volatilities for all equity variables are estimated from univariate GARCH(1,1) models; for the VIX we use the index level directly, as is standard in the volatility spillover literature.

股票市场变量：. 我们由美国股票构造四个板块组合（等权重、日度对数收益）：(i) 银行组合，包括摩根大通（JPM）、美国银行（BAC）、富国银行（WFC）、花旗集团（C）、高盛（GS）和摩根士丹利（MS）；(ii) 交易所组合，包括 Coinbase（COIN）和 Robinhood（HOOD）；(iii) 支付组合，包括 PayPal（PYPL）、Block（XYZ，原 Square）、Visa（V）和万事达（MA）；(iv) 加密货币概念组合，包括 MicroStrategy（MSTR）、Riot Platforms（RIOT）、Marathon Digital（MARA）、Hut 8（HUT）和 CleanSpark（CLSK）。所有股票数据来自 Yahoo Finance。我们还纳入标准普尔 500 指数（SPX）作为大盘基准，以及 CBOE VIX 指数作为市场不确定性的度量。所有股票变量的条件波动率均由单变量 GARCH(1,1) 模型估计；对于 VIX，按照波动溢出文献的标准做法，我们直接使用指数水平值。

State variables for tail risk regressions:. We include standard macrofinancial state variables from the Federal Reserve Economic Data (FRED) database: the 3-month Treasury bill rate, the term spread (10-year Treasury minus 3-month Treasury), the credit spread (Baa corporate bond yield minus 10-year Treasury yield), and the S&P 500 index return.

尾部风险回归的状态变量：. 我们纳入来自美联储经济数据（FRED）数据库的标准宏观金融状态变量：3 个月期国库券利率、期限利差（10 年期国债减 3 个月期国债）、信用利差（Baa 级公司债收益率减 10 年期国债收益率）以及标准普尔 500 指数收益。

Spillover system:. The TVP-VAR system contains eight variables: the conditional volatilities of USDT, USDC, the S&P 500, and the four sector portfolios, plus the VIX level. Because volatility series are highly persistent and fat-tailed, we transform each series to logarithms and winsorize at the 1% and 99% quantiles before estimation, which stabilizes the time-varying covariance estimates in the presence of extreme depegging observations.

溢出系统设定：. TVP-VAR 系统包含八个变量：USDT、USDC、标准普尔 500 指数和四个板块组合的条件波动率，外加 VIX 水平值。由于波动率序列高度持久且厚尾，我们在估计前对每个序列取对数并在 1% 与 99% 分位数处进行缩尾处理，以在存在极端脱锚观测的情况下稳定时变协方差估计。

Figure 1 plots the cumulative log returns of the five equity aggregates together with the two stablecoin market variables, with vertical dashed lines marking the key events of our sample: the Terra-LUNA collapse (May 9, 2022), the FTX bankruptcy (November 11, 2022), the SVB failure and USDC depegging (March 10, 2023), the Senate passage of the GENIUS Act (June 17, 2025), and its signing into law (July 18, 2025).

图 1 绘制了五个股票聚合序列的累计对数收益以及两个稳定币市场变量，竖直虚线标注了样本期内的关键事件：Terra-LUNA 崩盘（2022 年 5 月 9 日）、FTX 破产（2022 年 11 月 11 日）、SVB 倒闭与 USDC 脱锚（2023 年 3 月 10 日）、《GENIUS 法案》参议院通过（2025 年 6 月 17 日）及其签署成为法律（2025 年 7 月 18 日）。

Table 1 presents summary statistics for all variables used in our analysis.

表 1 给出了本文所用全部变量的描述性统计。

Several observations are worth noting. First, crypto-concept stocks and exchange stocks exhibit much higher volatility than banks, payment companies, or the broad market, with average conditional volatilities of 5.47% and 4.39% per day, respectively, compared to 1.54% for banks and 0.99% for the S&P 500. Second, changes in stablecoin market capitalization are positively skewed and strongly leptokurtic (kurtosis of 27.5), reflecting episodic large issuance and redemption events; the stablecoin basket price return is tightly centered at zero but exhibits occasional extreme negative realizations (down to −0.26% per day), corresponding to depegging episodes. Third, stablecoin price volatilities are orders of magnitude smaller than equity volatilities in normal times (means of 0.042% and 0.025% for USDT and USDC) but spike dramatically during depegging events (maxima of 0.58% and 0.47%), which is precisely the source of identifying variation for our spillover and tail risk analysis.

有几点观察值得注意。第一，加密货币概念股和交易所股票的波动率远高于银行、支付公司或大盘，日均条件波动率分别为 5.47% 和 4.39%，而银行为 1.54%、标准普尔 500 指数为 0.99%。第二，稳定币市值变化呈正偏且高度尖峰（峰度 27.5），反映了间歇性的大额增发与赎回事件；稳定币篮子价格收益紧密围绕零分布，但偶有极端负值（低至每日 −0.26%），对应脱锚事件。第三，稳定币价格波动率在正常时期比股票波动率低若干个数量级（USDT 和 USDC 的均值分别为 0.042% 和 0.025%），但在脱锚事件期间急剧飙升（最大值分别为 0.58% 和 0.47%），这正是本文溢出与尾部风险分析的识别变异来源。

Table 2 reports the unconditional correlation matrix of daily returns. The stablecoin quantity variable (ΔSC) is essentially uncorrelated with equity returns (correlations between

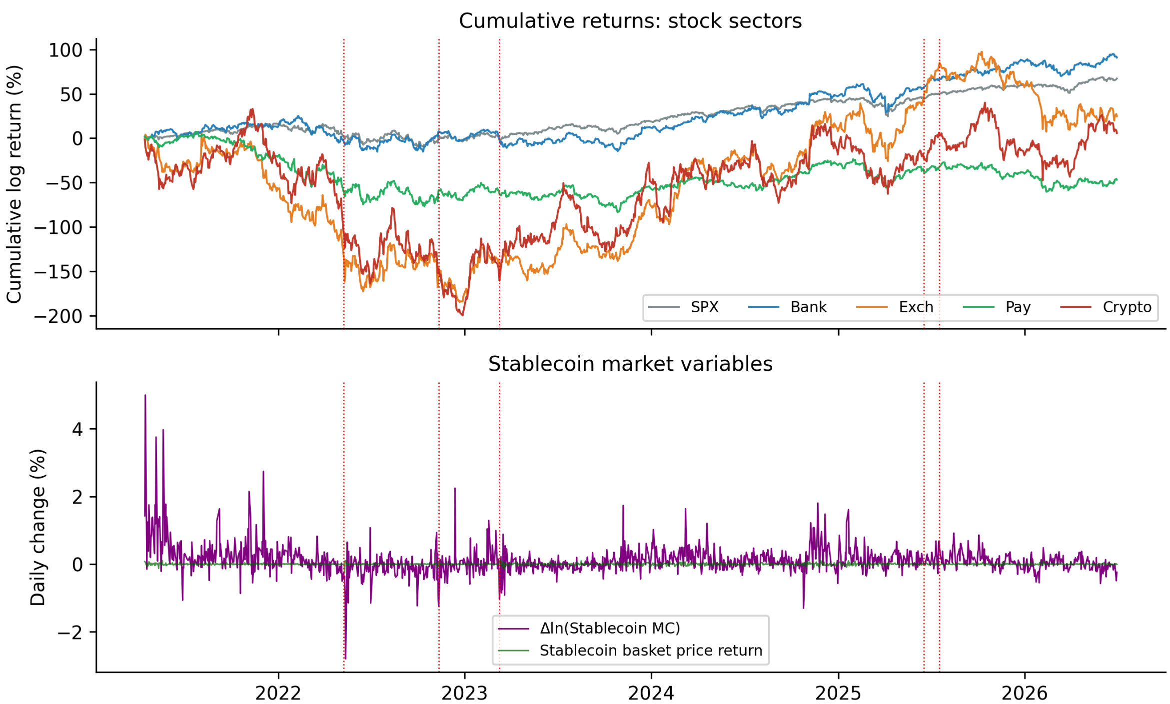

Figure 1: 累计收益与稳定币市场变量 / Cumulative returns and stablecoin market variables. The top panel shows cumulative daily log returns of the S&P 500 and the four sector portfolios; the bottom panel shows daily changes in stablecoin market capitalization and the stablecoin basket price return. Vertical dashed lines mark major crisis and regulatory events. / 上图显示标准普尔 500 指数与四个板块组合的累计日度对数收益；下图显示稳定币市值日度变化与稳定币篮子价格收益。竖直虚线标注主要危机与监管事件。

Table 1: 描述性统计 / Summary statistics

	Mean	Median	Std Dev	Min	Max	Skewness	Kurtosis
Panel A: Daily returns (%)							
Δln(Stablecoin MC)	0.112	0.056	0.424	-2.798	4.996	2.903	27.462
Stablecoin basket return	-0.000	-0.001	0.026	-0.258	0.115	-0.642	11.104
S&P 500	0.052	0.087	1.059	-6.161	9.089	0.012	6.473
Bank portfolio	0.071	0.071	1.559	-10.184	10.452	-0.056	4.523
Exchange portfolio	0.019	-0.105	4.462	-21.765	23.397	0.163	2.387
Payment portfolio	-0.036	0.083	1.908	-9.242	9.701	-0.156	2.497
Crypto-concept portfolio	0.004	-0.149	5.496	-21.113	22.695	0.195	0.675
Δln(VIX)	-0.166	-0.792	7.505	-44.245	55.411	0.831	6.452
Panel B: Conditional volatilities (%)							
USDT volatility	0.042	0.035	0.024	0.032	0.583	12.676	238.918
USDC volatility	0.025	0.022	0.025	0.016	0.469	13.190	190.420
S&P 500 volatility	0.992	0.887	0.357	0.591	3.702	2.473	10.214
Bank volatility	1.535	1.442	0.332	1.207	4.671	3.717	22.273
Exchange volatility	4.393	4.203	0.796	3.414	9.573	2.292	7.721
Payment volatility	1.826	1.653	0.513	1.192	3.476	1.449	1.361
Crypto-concept volatility	5.467	5.416	0.598	4.318	7.556	0.507	-0.129
VIX level	19.174	17.895	5.167	11.860	52.330	1.340	2.708

Notes: This table reports summary statistics for daily returns (Panel A) and conditional volatilities (Panel B) over the sample period April 15, 2021 to June 30, 2026 (1,288 trading days). Sector portfolios are equal-weighted. Conditional volatilities are estimated from univariate GARCH(1,1) models; the VIX enters in levels. 注：本表报告 2021 年 4 月 15 日至 2026 年 6 月 30 日样本期（1,288 个交易日）内日度收益（Panel A）与条件波动率（Panel B）的描述性统计。板块组合为等权重；条件波动率由单变量 GARCH(1,1) 模型估计；VIX 以水平值进入。

0.02 and 0.09), while the stablecoin basket price return (SCpx), which captures depegging pressure, correlates more visibly with crypto-concept stocks (0.277), exchange stocks (0.261), payment stocks (0.182), and the S&P 500 (0.157), and negatively with VIX changes (−0.130). Within equities, the S&P 500 is strongly correlated with payment stocks (0.754) and banks (0.712), exchange and crypto-concept stocks are highly correlated with each other (0.739), and VIX changes are negatively correlated with all equity returns, as expected.

表 2 报告了日度收益的无条件相关系数矩阵。稳定币数量变量（ΔSC）与股票收益基本不相关（相关系数介于 0.02 至 0.09 之间），而捕捉脱锚压力的稳定币篮子价格收益（SCpx）与加密货币概念股（0.277）、交易所股票（0.261）、支付股票（0.182）和标准普尔 500 指数（0.157）的相关性更为明显，并与 VIX 变动负相关（−0.130）。在股票内部，标准普尔 500 指数与支付股（0.754）和银行股（0.712）高度相关，交易所股票与加密货币概念股之间高度相关（0.739），VIX 变动与所有股票收益负相关，符合预期。

Table 2: 日度收益的无条件相关矩阵 / Unconditional correlation matrix of daily returns

	ΔSC	SCpx	SPX	Bank	Exch	Pay	Crypto	ΔVIX
ΔSC	1.00							
SCpx	0.14	1.00						
SPX	0.05	0.16	1.00					
Bank	0.06	0.09	0.71	1.00				
Exch	0.05	0.26	0.57	0.45	1.00			
Pay	0.02	0.18	0.75	0.57	0.57	1.00		
Crypto	0.09	0.28	0.56	0.42	0.74	0.52	1.00	
ΔVIX	-0.08	-0.13	-0.77	-0.60	-0.46	-0.57	-0.46	1.00

Notes: This table reports Pearson correlation coefficients for daily returns. ΔSC denotes daily log changes in total stablecoin market capitalization; SCpx denotes the supply-weighted stablecoin basket price return. 注：本表报告日度收益的皮尔逊相关系数。ΔSC 表示稳定币总市值的日度对数变化；SCpx 表示供给加权的稳定币篮子价格收益。

5.2. Stationarity and model specification / 平稳性与模型设定

Before estimating the TVP-VAR model, we conduct standard unit root tests (Augmented Dickey-Fuller) on all variables. The test results (available upon request) strongly reject the null hypothesis of a unit root for all return series and log-transformed volatility series, confirming that all variables are stationary and suitable for VAR estimation. We select a lag length of $p = 2$ based on the Bayesian Information Criterion (BIC), and we use a forecast horizon of $H = 10$ days for our baseline spillover estimates; Section 8 shows that our results are robust to alternative lag lengths and forecast horizons.

在估计 TVP-VAR 模型之前，我们对所有变量进行标准的单位根检验（增广 Dickey-Fuller 检验）。检验结果（备索）强烈拒绝所有收益序列与对数变换波动率序列存在单位根的原假设，确认所有变量平稳、适用于 VAR 估计。我们基于贝叶斯信息准则（BIC）选择滞后阶数 $p = 2$，基准溢出估计使用 $H = 10$ 天的预测期限；第 8 节表明我们的结果对不同的滞后阶数与预测期限保持稳健。

6. Empirical results / 实证结果

6.1. Static spillover analysis / 静态溢出分析

We begin by discussing the average (static) spillover measures, which are obtained by averaging the time-varying spillover indices over the full sample. Table 3 reports the full spillover table, and Figure 2 visualizes the average spillover matrix as a heatmap.

我们首先讨论平均（静态）溢出测度，即对全样本期内时变溢出指数取平均。表 3 报告了完整的溢出表，图 2 以热力图形式可视化平均溢出矩阵。

Table 3: 静态波动溢出表（全样本平均）/ Static volatility spillover table (full-sample averages)

	USDT	USDC	SPX	Bank	Exch	Pay	Crypto	VIX	FROM others
USDT	**62.86**	14.71	1.92	2.27	8.11	2.95	3.72	3.46	37.14
USDC	7.98	**71.87**	1.87	2.92	6.69	3.57	1.59	3.51	28.13
SPX	6.52	8.98	**27.01**	9.47	4.34	8.35	2.92	32.42	72.99
Bank	5.08	5.85	10.55	**50.85**	5.14	6.44	3.11	12.99	49.15
Exch	6.44	12.29	3.58	5.34	**48.25**	6.92	12.50	4.67	51.75
Pay	4.17	5.22	12.65	6.29	6.35	**48.68**	3.70	12.94	51.32
Crypto	8.66	11.12	4.16	5.21	14.53	6.12	**44.68**	5.51	55.32
VIX	6.97	10.85	1.80	3.19	4.81	2.20	1.70	**68.47**	31.53
TO others	45.83	69.01	36.52	34.69	49.97	36.55	29.23	75.49	**47.16**
NET spillovers	8.70	40.89	-36.47	-14.46	-1.78	-14.76	-26.08	43.97	

Notes: This table reports the full-sample average volatility spillover indices based on the TVP-VAR model with 2 lags, forgetting factors $\kappa_1 = \kappa_2 = 0.99$, and 10-step-ahead generalized forecast error variance decomposition of log-volatilities. The i, j-th entry gives the estimated contribution to the forecast error variance of variable i coming from shocks to variable j; diagonal entries (own shares) are in bold. The "FROM others" column sums all off-diagonal entries in each row, the "TO others" row sums all off-diagonal entries in each column, and "NET spillovers" is the difference between TO and FROM. The total spillover index (bottom-right entry) equals 47.16%. All entries are in percentage points. 注：本表报告基于 TVP-VAR 模型（2 阶滞后、遗忘因子 $\kappa_1 = \kappa_2 = 0.99$、对数波动率的 10 步前向广义预测误差方差分解）的全样本平均波动溢出指数。第 i 行第 j 列元素表示来自变量 j 的冲击对变量 i 预测误差方差的估计贡献；对角线元素（自身份额）以粗体显示。总溢出指数（右下角）为 47.16%。所有元素均以百分点表示。

The total connectedness index is 47.16%, indicating that on average nearly half of the forecast error variance across the system comes from cross-variable spillovers rather than own-variable shocks. This level of connectedness is substantially higher than typical estimates for stablecoin-equity systems obtained from returns-based measures, reflecting the fact that our system includes highly integrated crypto-exposed equities (exchanges and crypto-concept stocks) whose volatilities share large common factors with the VIX and with each other.

总关联性指数为 47.16%，表明平均而言，系统中近一半的预测误差方差来自跨变量溢出而非自身冲击。这一关联水平明显高于基于收益测度的稳定币—股票系统典型估计值，这反映了我们的系统包含了高度一体化的加密暴露股票（交易所与加密货币概念股），其波动率与 VIX 以及彼此之间共享着巨大的共同因子。

Looking at directional spillovers, the VIX is the largest net transmitter of volatility in the system (NET = +44.0), followed by USDC (+40.9) and USDT (+8.7): both stablecoins

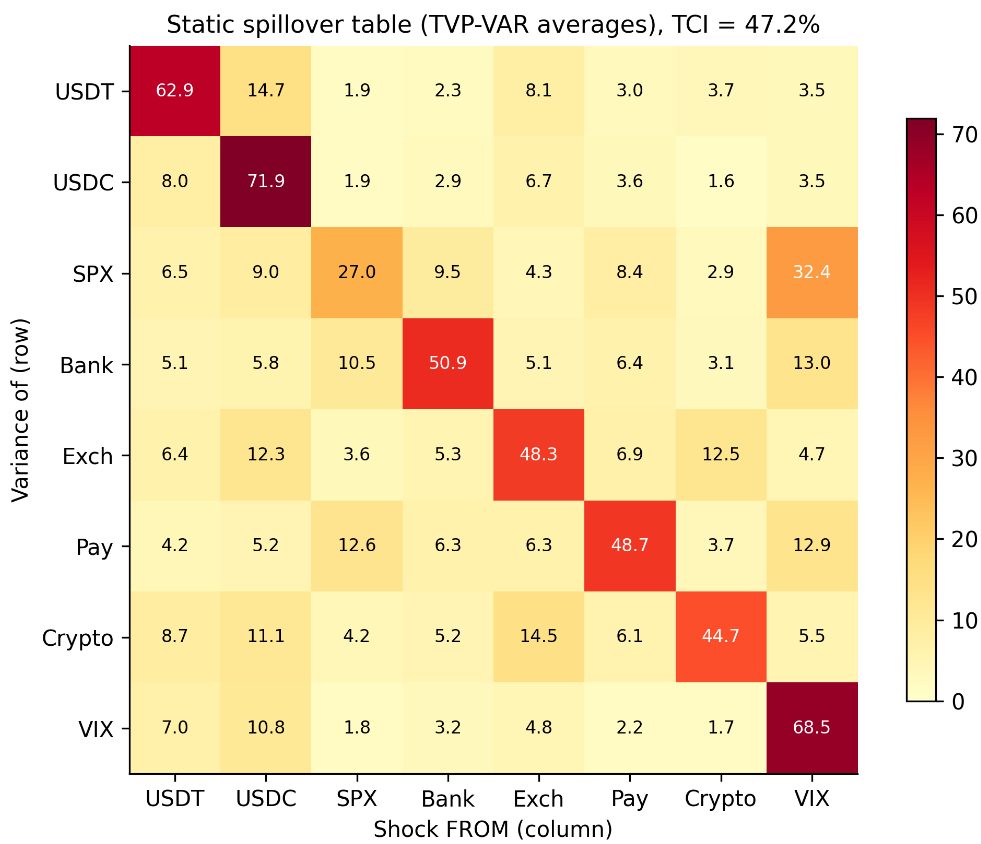

Figure 2: 静态溢出矩阵热力图 / Heatmap of the static spillover table (TVP-VAR full-sample averages), TCI = 47.2%. / 静态溢出表（TVP-VAR 全样本平均）热力图，总溢出指数 = 47.2%。

are net transmitters of volatility to the rest of the system on average. The S&P 500 is by far the largest net receiver (NET = −36.5), with the VIX alone accounting for 32.4% of the S&P 500's forecast error variance. Crypto-concept stocks (−26.1), payment stocks (−14.8), and bank stocks (−14.5) are also net receivers, while exchange stocks are approximately neutral (−1.8). The own-variable shares are highest for USDC (71.9%) and the VIX (68.5%) and lowest for the S&P 500 (27.0%), indicating that broad-market volatility is the most "system-driven" variable in the network.

从方向性溢出来看，VIX 是系统中最大的波动净输出者（NET = +44.0），其次是 USDC（+40.9）和 USDT（+8.7）：两种稳定币平均而言都是对系统其余部分的波动净输出者。标准普尔 500 指数是迄今最大的净接收者（NET = −36.5），仅 VIX 就贡献了其预测误差方差的 32.4%。加密货币概念股（−26.1）、支付股（−14.8）和银行股（−14.5）也是净接收者，而交易所股票大致中性（−1.8）。自身份额最高的是 USDC（71.9%）和 VIX（68.5%），最低的是标准普尔 500 指数（27.0%），表明大盘波动率是网络中最"受系统驱动"的变量。

Examining bilateral linkages, USDT and USDC are strongly interconnected with each other (14.7% and 8.0% pairwise contributions), and both transmit meaningful volatility to exchange stocks (8.1% and 6.7%) and crypto-concept stocks (3.7% and 1.6%). Exchange stocks transmit 12.5% of crypto-concept stocks' variance, and crypto-concept stocks transmit 14.5% back, confirming strong bidirectional connectedness within the crypto ecosystem. Direct stablecoin-to-bank spillovers are modest on average (2.3% for USDT and 2.9% for USDC), but—as the dynamic analysis below shows—they intensify sharply during crisis episodes.

考察双边联系，USDT 与 USDC 彼此高度互联（两两贡献分别为 14.7% 和 8.0%），且两者都向交易所股票（8.1% 和 6.7%）与加密货币概念股（3.7% 和 1.6%）输出可观的波动。交易所股票贡献了加密货币概念股方差的 12.5%，加密货币概念股则反向输出 14.5%，证实了加密生态系统内部强的双向关联性。稳定币对银行的直接溢出平均而言不大（USDT 为 2.3%、USDC 为 2.9%），但正如下文的动态分析所示，它们在危机期间会急剧增强。

6.2. *Time-varying spillover dynamics* / 时变溢出动态

While the average spillover index is informative, the key advantage of the TVP-VAR approach is its ability to capture how connectedness evolves over time. Figure 3 plots the dynamic total connectedness index, with vertical lines marking major events; Figure 4 plots the time-varying net directional spillovers of each variable.

平均溢出指数虽然提供了信息，但 TVP-VAR 方法的关键优势在于捕捉关联性如何随时间演化。图 3 绘制了动态总关联性指数，竖线标注主要事件；图 4 绘制了各变量的时变净方向性溢出。

The total spillover index shows dramatic time variation, ranging from a minimum of 29.9% (July 2024) to a maximum of 90.0%. We identify four key episodes:

总溢出指数表现出剧烈的时变性，从最低 29.9%（2024 年 7 月）到最高 90.0% 不等。我们识别出四个关键阶段：

1. **Terra-LUNA collapse (May 2022):** The TCI rose from around 33% to a local peak of 47.4% in the days following the TerraUSD depegging, with the ±30-trading-day event-window average reaching only 35.0%. Spillovers from the crypto-native crisis to the broader system were therefore material but contained.

1. **Terra-LUNA 崩盘（2022 年 5 月）：**TCI 从约 33% 上升至 TerraUSD 脱锚后数日内的局部峰值 47.4%，事件前后 30 个交易日窗口的均值仅为 35.0%。因此，加密原生危机

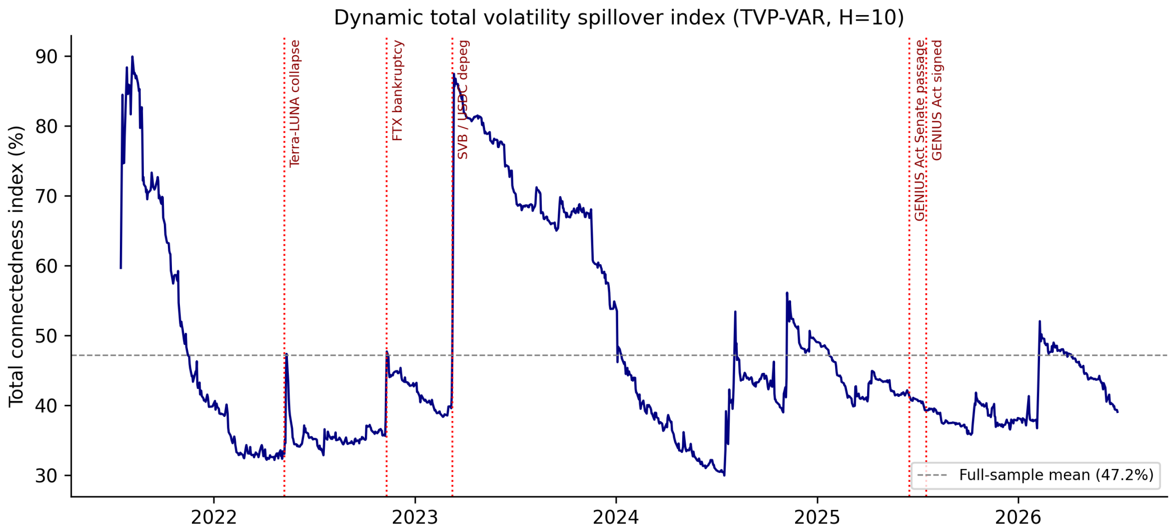

Figure 3: 动态总波动溢出指数 / Dynamic total volatility spillover index (TVP-VAR, $H = 10$). Vertical lines mark the Terra-LUNA collapse (May 2022), the FTX bankruptcy (November 2022), the SVB/USDC depegging (March 2023), the Senate passage of the GENIUS Act (June 2025), and its signing into law (July 2025). / 动态总波动溢出指数（TVP-VAR，$H = 10$）。竖线标注 Terra-LUNA 崩盘（2022 年 5 月）、FTX 破产（2022 年 11 月）、SVB/USDC 脱锚（2023 年 3 月）、《GENIUS 法案》参议院通过（2025 年 6 月）及其签署成为法律（2025 年 7 月）。

对更广泛系统的溢出是实质性的但相对受控。

2. **FTX bankruptcy (November 2022):** The TCI climbed again to a peak of 47.7% around FTX's Chapter 11 filing, with an event-window average of 40.5%, as spillovers from crypto-concept and exchange stocks to the broader financial system intensified.

2. **FTX 破产（2022 年 11 月）**：TCI 在 FTX 申请第十一章破产保护前后再次攀升至 47.7% 的峰值，事件窗口均值为 40.5%，来自加密货币概念股和交易所股票对更广泛金融体系的溢出有所增强。

3. **SVB failure and USDC depegging (March 2023):** This episode produced by far the largest spillover spike in the sample. The TCI jumped from 41.3% on March 9, 2023 to 87.5% on March 13, 2023—the first trading day after USDC lost its peg amid concerns over Circle's \$3.3 billion reserve exposure to SVB—and remained above 80% for several months. The ±30-day event-window average reached 60.2%, nearly double the Terra episode. This pattern shows that a stablecoin crisis originating in the traditional banking system, which directly threatened the perceived safety of stablecoin reserves, generated far stronger economy-wide connectedness than crypto-native shocks.

3. **SVB 倒闭与 USDC 脱锚（2023 年 3 月）**：这一阶段产生了样本中迄今最大的溢出飙升。TCI 从 2023 年 3 月 9 日的 41.3% 跳升至 3 月 13 日的 87.5%——这是市场担忧 Circle 在 SVB 的 33 亿美元储备敞口导致 USDC 脱锚后的首个交易日——并在随后数月内保持在 80% 以上。事件前后 30 天窗口均值达到 60.2%，几乎是 Terra 事件的两倍。这一模式表明，一场源自传统银行体系、直接威胁稳定币储备安全性的稳定币危机，所产生的全系统关联性远强于加密原生冲击。

4. **GENIUS Act legislative process (June–July 2025):** In contrast to the crisis episodes, the Senate passage of the GENIUS Act (June 17, 2025) and its signing into law

(July 18, 2025) were associated with event-window averages of 41.1% and 39.7%, close to the sample mean and without any pronounced spike. This suggests that the market had largely anticipated the legislation, and that regulatory clarity per se does not amplify volatility connectedness.

4. **《GENIUS 法案》立法进程（2025 年 6–7 月）**：与危机阶段形成鲜明对比的是，《GENIUS 法案》参议院通过（2025 年 6 月 17 日）和签署成为法律（2025 年 7 月 18 日）对应的事件窗口均值分别为 41.1% 和 39.7%，接近样本均值且无明显飙升。这表明市场已基本预期到该立法，监管明晰本身并不会放大波动关联性。

These results provide strong support for Hypothesis H1: connectedness between stablecoin markets and equity markets is highly time-varying and intensifies dramatically during crisis events—most powerfully when the crisis directly involves the stablecoin peg itself.

这些结果为假说 H1 提供了有力支持：稳定币市场与股票市场之间的关联性高度时变，并在危机事件期间急剧增强——尤其是当危机直接涉及稳定币锚定本身时。

The time-varying net spillovers in Figure 4 reveal additional structure. The VIX is a persistent net transmitter, and its net transmission strengthens precisely during equity stress episodes (most visibly in early 2022 and late 2024). USDC's net transmission surges dramatically during the March 2023 depegging and remains elevated for over a year, consistent with the lasting repricing of stablecoin reserve risk after the SVB episode. The S&P 500 is a persistent net receiver, absorbing volatility from all corners of the system.

图 4 中的时变净溢出揭示了更多结构。VIX 是持续的净输出者，且其净输出恰在股市承压阶段增强（最明显的是 2022 年初和 2024 年末）。USDC 的净输出在 2023 年 3 月脱锚期间急剧飙升，并在随后一年多的时间里保持高位，与 SVB 事件后稳定币储备风险的持久重定价相一致。标准普尔 500 指数是持续的净接收者，吸收来自系统各方的波动。

6.3. Frequency-domain decomposition / 频域分解

To examine whether spillovers are driven by short-term trading/sentiment or long-term fundamental linkages (Hypothesis H3), we present the frequency decomposition of spillovers in Table 4 and Figure 5.

为考察溢出是由短期交易/情绪驱动还是由长期基本面联系驱动（假说 H3），我们在表 4 和图 5 中给出溢出的频域分解。

Contrary to Hypothesis H3, the results show that volatility spillovers are dominated by the low-frequency component: the long-term band (21+ days) accounts for 48.6% of total spillovers, the medium-term band for 31.4%, and the short-term band (1–5 days) for only 20.0%. The within-band total spillover index rises monotonically from 29.3% at short horizons to 71.0% at long horizons. Because the analysis is conducted on (log) conditional volatilities—which are themselves highly persistent, slow-moving objects—the dominance of low-frequency components indicates that cross-market linkages operate through sustained volatility regimes rather than through transient high-frequency co-movement. In economic terms, the connectedness we document is not primarily the product of day-to-day sentiment spillovers or algorithmic trading, but reflects persistent shared exposure to common risk factors (market stress, liquidity conditions, and crypto-ecosystem risk).

与假说 H3 相反，结果显示波动溢出由低频成分主导：长期频段（21 天以上）占总溢出的 48.6%，中期频段占 31.4%，短期频段（1–5 天）仅占 20.0%。频段内总溢出指数从短期限的 29.3% 单调上升至长期限的 71.0%。由于分析对象是（对数）条件波动率——其本身就是高度持久、缓慢变动的对象——低频成分的主导地位表明，跨市场联系通过持续的

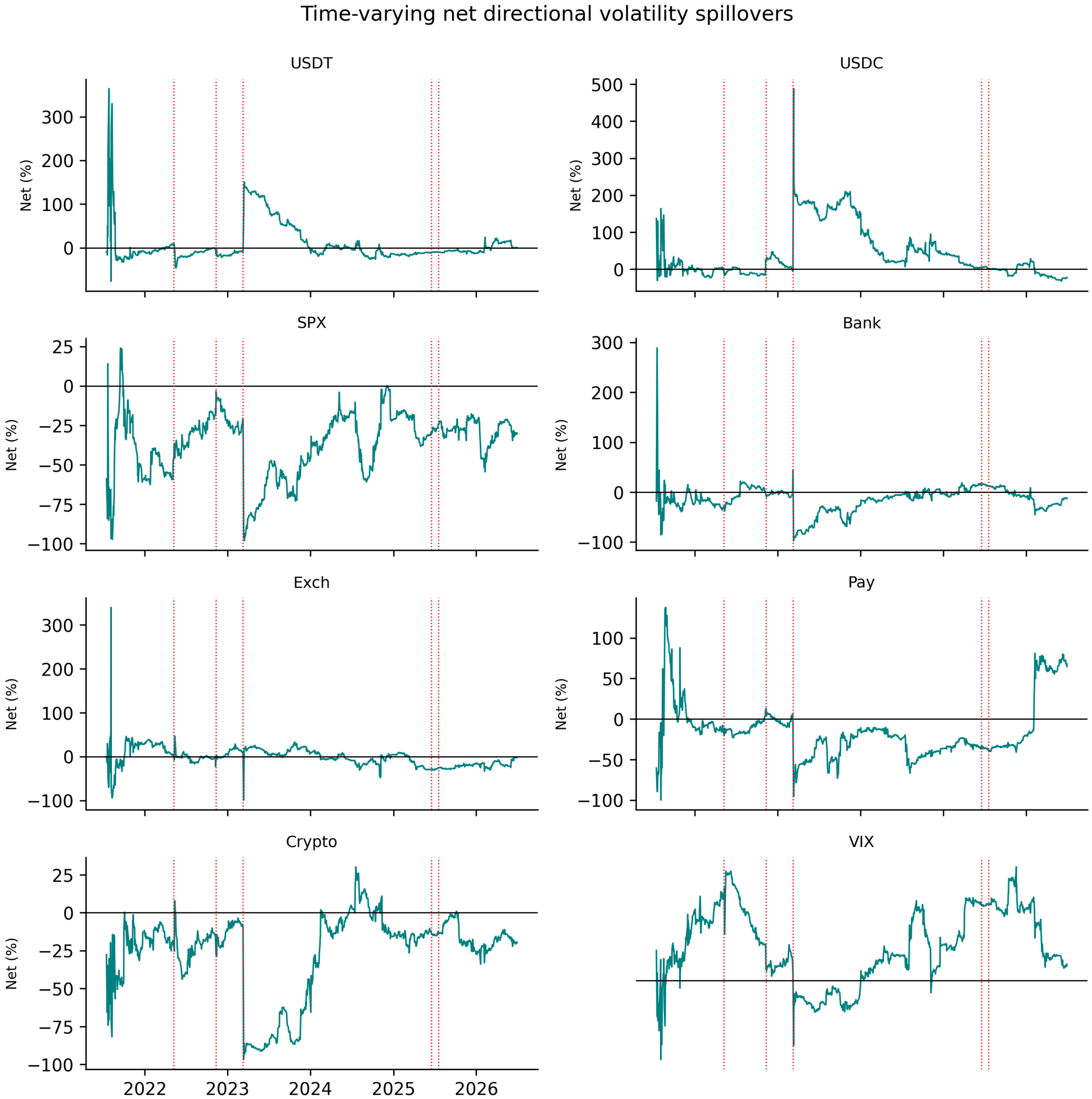

Figure 4: 时变净方向性波动溢出 / Time-varying net directional volatility spillovers. Positive values indicate net transmission of volatility to the rest of the system. / 时变净方向性波动溢出。正值表示对系统其余部分的波动净输出。

Table 4: 波动溢出的频域分解 / Frequency-domain decomposition of volatility spillovers

Frequency band	Short-term 1–5 days	Medium-term 6–20 days	Long-term 21+ days
Total spillover (%)	29.28	45.85	71.04
% of total spillovers	20.03%	31.37%	48.60%
Net spillovers by variable:			
USDT	12.11	-51.71	-94.30
USDC	-6.85	22.14	-61.45
S&P 500	19.94	22.30	58.45
Banks	3.59	-16.19	-79.31
Exchanges	-3.05	-1.73	-56.97
Payments	-10.81	-0.01	64.54
Crypto-concept	-5.92	26.59	201.50
VIX	-9.02	-1.40	-32.46

Notes: This table decomposes the total volatility spillover index into frequency bands following Baruník and Křehlík [5]. The short-term band corresponds to 1–5 days, medium-term to 6–20 days, and long-term to horizons beyond 21 days. The first row reports the within-band total spillover index; the second row reports each band's share in the sum of within-band spillovers; the lower panel reports within-band net spillovers. All entries are in percentage points. 注：本表按照 Baruník and Křehlík [5] 的方法将总波动溢出指数分解至各频段。短期频段对应 1–5 天，中期对应 6–20 天，长期对应 21 天以上。第一行为频段内总溢出指数；第二行为各频段溢出占总溢出的份额；下半部分为频段内净溢出。所有元素均以百分点表示。

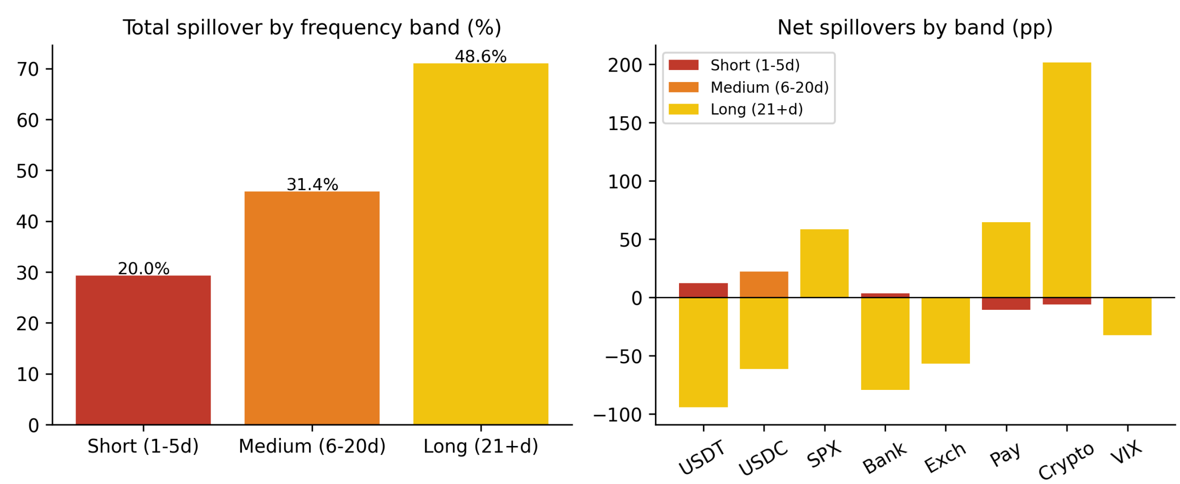

Figure 5: 溢出的频域分解 / Frequency decomposition of volatility spillovers. Left panel: total spillover by frequency band; right panel: net spillovers by band. / 波动溢出的频域分解。左图：分频段总溢出；右图：分频段净溢出。

波动状态机制发挥作用，而非短暂的高频联动。从经济含义上看，我们所记录的关联性主要不是日间情绪溢出或算法交易的产物，而是反映了对共同风险因子（市场压力、流动性状况和加密生态系统风险）的持久共享暴露。

The frequency-specific net spillovers are also informative. Crypto-concept stocks are by far the largest net receivers at the long horizon (+201.5 in absolute net inflow), and the S&P 500 and payment stocks are large long-horizon receivers as well, whereas the stablecoins USDT and USDC are strong long-horizon net transmitters (−94.3 and −61.5 on the receiver side of the ledger). At short horizons, the S&P 500 and USDT are the main net transmitters, consistent with broad-market news and stablecoin supply shocks propagating quickly.

分频段净溢出同样具有信息量。加密货币概念股是长期频段内迄今最大的净接收方（绝对净流入 +201.5），标准普尔 500 指数与支付股也是长期频段的大型净接收方，而稳定币 USDT 和 USDC 是长期频段的强净输出方（接收侧为 −94.3 和 −61.5）。在短期频段，标准普尔 500 指数和 USDT 是主要的净输出者，与大盘消息和稳定币供给冲击的快速传播相一致。

6.4. *Extreme risk contagion: CoVaR and MES results* / 极端风险传染：*CoVaR* 与 *MES* 结果

We now turn to tail risk analysis to test Hypothesis H4. Table 5 reports the 5% ΔCoVaR and MES estimates for each stock sector conditional on stablecoin market distress (the supply-weighted basket price return at its 5% quantile), along with block-bootstrap standard errors. Figures 6 and 7 display the estimates with 95% confidence intervals.

我们现在转向尾部风险分析以检验假说 H4。表 5 报告了以稳定币市场危机（供给加权篮子价格收益处于其 5% 分位数）为条件的各股票板块 5% ΔCoVaR 与 MES 估计值及区组自助法标准误。图 6 和图 7 展示了估计值及其 95% 置信区间。

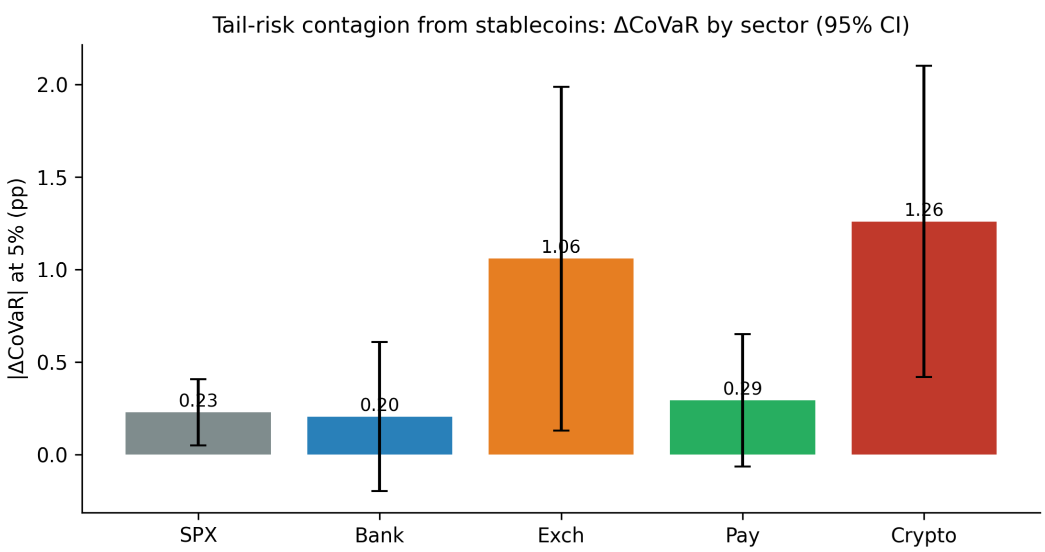

Figure 6: 稳定币对各板块的尾部风险传染（ΔCoVaR）/ Tail-risk contagion from stablecoins: ΔCoVaR by sector (95% confidence intervals). / 稳定币的尾部风险传染：分板块 ΔCoVaR（95% 置信区间）。

The results provide strong support for Hypotheses H2 and H4 and reveal dramatic heterogeneity across sectors. Crypto-concept stocks exhibit the largest tail exposure: when

Table 5: 尾部风险传染：ΔCoVaR 与 MES 估计 / Tail risk contagion: ΔCoVaR and MES estimates

Sector	VaR (5%)	ΔCoVaR (5%)	MES (5%)
S&P 500	-1.631%	-0.229%**	-0.173%
		(0.091)	(0.157)
Bank portfolio	-2.409%	0.205%	-0.050%
		(0.206)	(0.180)
Exchange portfolio	-6.904%	-1.058%**	-1.881%***
		(0.474)	(0.660)
Payment portfolio	-3.391%	-0.292%	-0.642%**
		(0.182)	(0.291)
Crypto-concept portfolio	-8.703%	-1.259%***	-2.536%***
		(0.429)	(0.694)
Tests for equality of $\Delta CoVaR$ across sectors:			
Crypto vs Bank		-1.463%***	
		(0.459)	
Crypto vs Payment		-0.966%**	
		(0.471)	
Crypto vs Exchange		-0.201%	
		(0.474)	

Notes: This table reports 5% Value-at-Risk (VaR), ΔCoVaR, and Marginal Expected Shortfall (MES) estimates conditional on the supply-weighted stablecoin basket price return being at its 5% quantile. ΔCoVaR measures the change in the sector's VaR when stablecoin markets are in distress relative to their median state. MES measures the expected sector return on days when the stablecoin basket is in its 5% left tail. All estimates are based on quantile regressions controlling for lagged VIX changes, S&P 500 returns, the term spread, the credit spread, and the 3-month T-bill rate. Block-bootstrap standard errors (200 replications) are in parentheses. ***, **, * denote statistical significance at the 1%, 5%, and 10% levels, respectively. 注：本表报告以供给加权稳定币篮子价格收益处于其 5% 分位数为条件的 5% 在险价值（VaR）、ΔCoVaR 和边际期望损失（MES）估计值。ΔCoVaR 衡量稳定币市场处于危机状态相对中位状态时板块 VaR 的变化；MES 衡量稳定币篮子处于 5% 左尾时板块的预期收益。所有估计均基于控制了滞后 VIX 变动、标准普尔 500 指数收益、期限利差、信用利差和 3 个月期国库券利率的分位数回归。括号内为区组自助法（200 次）标准误。***、**、* 分别表示在 1%、5% 和 10% 水平上显著。

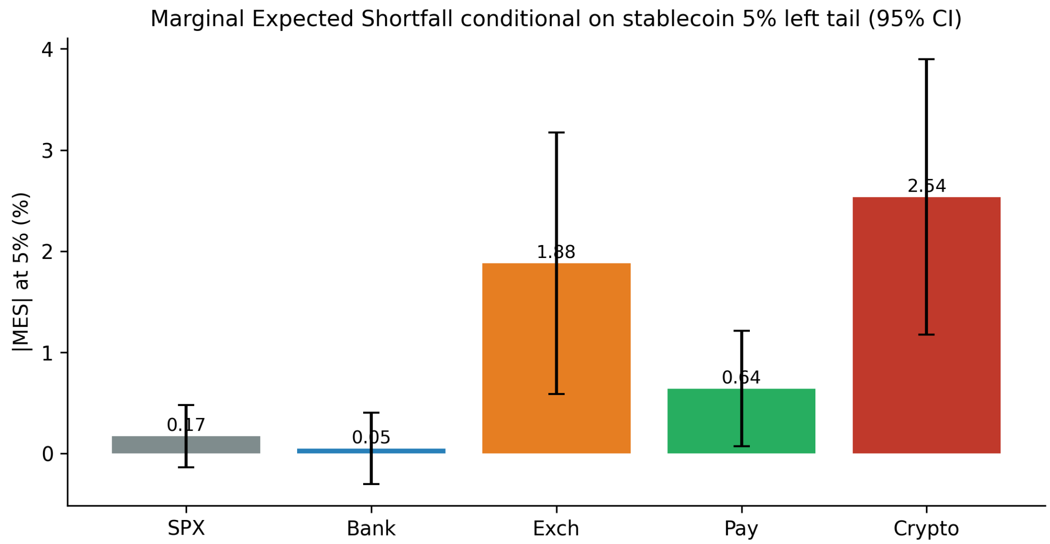

Figure 7: 以稳定币 5% 左尾为条件的边际期望损失 / Marginal Expected Shortfall conditional on the stablecoin 5% left tail (95% confidence intervals). / 以稳定币 5% 左尾为条件的 MES（95% 置信区间）。

stablecoin markets are in distress, their 5% VaR worsens by an additional 1.26 percentage points (ΔCoVaR = −1.259%, significant at the 1% level), and their expected loss on the worst 5% of stablecoin days is −2.54% (MES, significant at 1%). Exchange stocks are the next most vulnerable (ΔCoVaR = −1.058%, significant at 5%; MES = −1.881%, significant at 1%). The S&P 500 shows a smaller but statistically significant ΔCoVaR of −0.229%, and payment stocks a marginally significant one (−0.292%, p = 0.109) with a significant MES of −0.642%. Most strikingly, bank stocks show no significant tail exposure at all: their ΔCoVaR estimate is small and positive (0.205%, insignificant) and their MES is essentially zero (−0.050%).

结果为假说 H2 和 H4 提供了有力支持，并揭示了板块间的巨大异质性。加密货币概念股的尾部暴露最大：当稳定币市场承压时，其 5% VaR 额外恶化 1.26 个百分点（ΔCoVaR = −1.259%，在 1% 水平显著），在稳定币最差 5% 交易日的预期损失为 −2.54%（MES，在 1% 水平显著）。交易所股票次之（ΔCoVaR = −1.058%，在 5% 水平显著；MES = −1.881%，在 1% 水平显著）。标准普尔 500 指数的 ΔCoVaR 较小但统计显著（−0.229%），支付股的 ΔCoVaR 边际显著（−0.292%，p = 0.109）且 MES 显著（−0.642%）。最引人注目的是，银行股完全不存在显著的尾部暴露：其 ΔCoVaR 估计值很小且为正（0.205%，不显著），MES 基本为零（−0.050%）。

The bottom panel of Table 5 formally tests whether these differences are statistically significant. The ΔCoVaR of crypto-concept stocks is 1.46 percentage points larger in magnitude than that of bank stocks (significant at 1%) and 0.97 percentage points larger than that of payment stocks (significant at 5%), while the difference between crypto-concept and exchange stocks is not statistically significant (p = 0.671). The sectoral ranking of tail exposure—crypto-concept > exchange > payment > broad market > banks—is exactly the ordering predicted by Hypothesis H2 and mirrors the sectors' fundamental exposure to the crypto ecosystem.

表 5 下半部分正式检验了这些差异的统计显著性。加密货币概念股的 ΔCoVaR 绝对值

比银行股大 1.46 个百分点（在 1% 水平显著），比支付股大 0.97 个百分点（在 5% 水平显著），而加密货币概念股与交易所股票之间的差异不显著（$p = 0.671$）。尾部暴露的板块排序——加密货币概念股 > 交易所 > 支付 > 大盘 > 银行——与假说 H2 预测的次序完全一致，并反映了各板块对加密生态系统的基本面敞口。

To understand how tail risk contagion varies over time, we also estimate ΔCoVaR recursively using a 250-day rolling window (Figure 8). Tail risk exposure of crypto-concept and exchange stocks spikes around the major crisis episodes identified in the spillover analysis: the rolling ΔCoVaR of crypto-concept stocks fluctuates around a full-sample median of roughly 1.4 percentage points in magnitude and peaks during the Terra-LUNA window, while remaining consistently larger than that of banks, whose rolling estimates hover near zero throughout.

为理解尾部风险传染如何随时间变化，我们还使用 250 天滚动窗口递归估计 ΔCoVaR（图 8）。加密货币概念股与交易所股票的尾部风险暴露在溢出分析所识别的主要危机阶段前后出现飙升：加密货币概念股的滚动 ΔCoVaR 围绕约 1.4 个百分点的全样本中位数（绝对值）波动，并在 Terra-LUNA 窗口达到峰值；而银行股的滚动估计值全程在零附近徘徊，始终远小于前者。

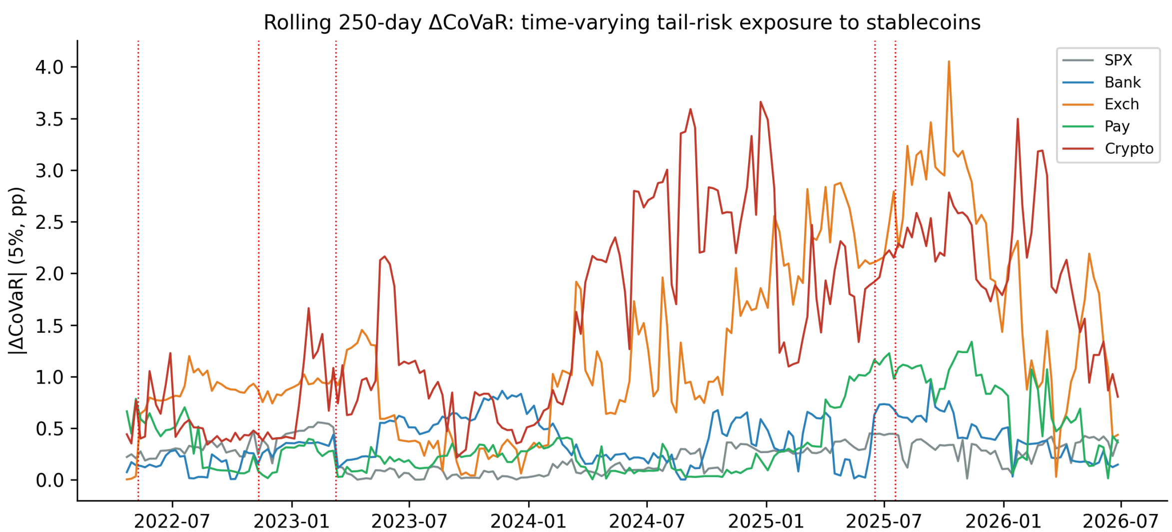

Figure 8: 滚动 250 天 ΔCoVaR / Rolling 250-day ΔCoVaR: time-varying tail-risk exposure to stablecoins. / 滚动 250 天 ΔCoVaR：对稳定币的时变尾部风险暴露。

7. Heterogeneity analysis / 异质性分析

7.1. Sector-level net spillovers / 板块层面的净溢出

To explore sectoral heterogeneity more deeply, Figure 9 presents the time-varying pairwise net spillovers between the stablecoin block (USDT and USDC combined) and each stock sector, and Figure 10 aggregates the stablecoin block's net spillover position vis-à-vis all equity variables and the VIX.

为更深入地考察板块异质性，图 9 展示了稳定币板块（USDT 与 USDC 合并）与各股票板块之间的时变两两净溢出，图 10 汇总了稳定币板块相对所有股票变量与 VIX 的净溢出头寸。

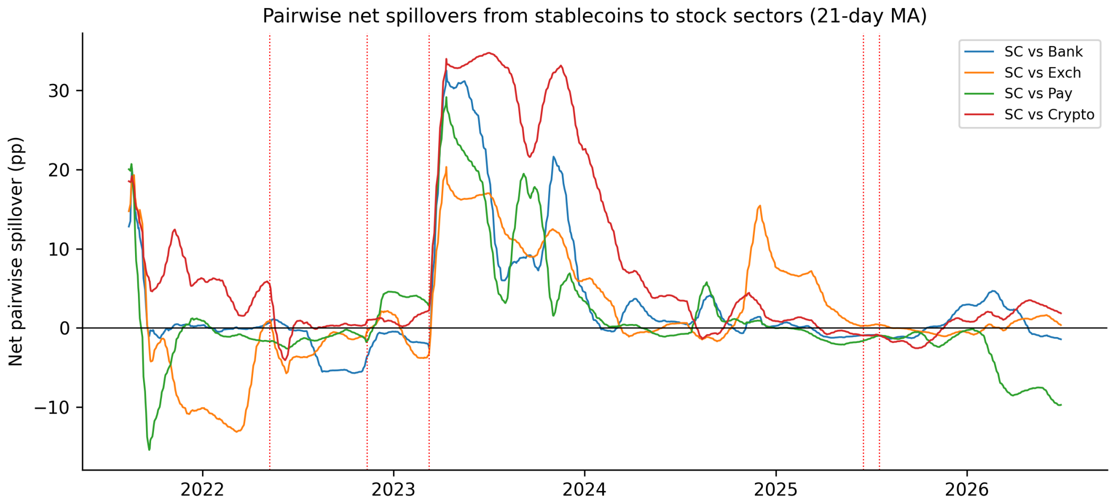

Figure 9: 稳定币对各股票板块的两两净溢出 / Pairwise net spillovers from stablecoins to stock sectors (21-day moving average). / 稳定币对股票板块的两两净溢出（21 日移动平均）。

The pairwise spillover dynamics reveal rich heterogeneity. Stablecoin-to-exchange and stablecoin-to-crypto spillovers are the largest in magnitude and highly time-varying, with stablecoins transmitting significant volatility to these sectors during crypto stress events and during the 2023 USDC depegging. Stablecoin-to-payment spillovers rose notably from 2023 onward, consistent with PayPal's launch of PYUSD and Visa's stablecoin settlement pilots deepening the link between payment incumbents and stablecoin infrastructure. Stablecoin-to-bank spillovers are generally close to zero outside of crisis windows.

两两溢出动态揭示了丰富的异质性。稳定币对交易所和稳定币对加密货币概念股的溢出幅度最大且高度时变，在加密压力事件和 2023 年 USDC 脱锚期间，稳定币向这些板块输出了显著的波动。稳定币对支付股的溢出从 2023 年起明显上升，与 PayPal 推出 PYUSD 以及 Visa 开展稳定币结算试点加深了支付巨头与稳定币基础设施之间联系的事实相一致。在危机窗口之外，稳定币对银行的溢出总体上接近于零。

Figure 10 shows that the stablecoin block is a net volatility transmitter vis-à-vis the equity sectors and the VIX throughout most of the sample. Averaged over event windows, the stablecoin block's net spillover is 48.9 percentage points during tranquil periods and rises to 57.3 percentage points during crisis windows. Contrary to the notion that stablecoins merely absorb shocks from traditional markets, they are a persistent source of volatility for equities in our system, and this transmitter role strengthens—rather than reverses—when the system is under stress.

图 10 显示，稳定币板块在样本期的大部分时间内都是相对股票板块和 VIX 的波动净输出者。按事件窗口平均，稳定币板块的净溢出在平静时期为 48.9 个百分点，在危机窗口上升至 57.3 个百分点。与"稳定币仅仅吸收来自传统市场的冲击"这一观念相反，在我们的系统中，稳定币是股票市场波动的持久来源，且这一输出者角色在系统承压时增强而非逆转。

7.2. Market state dependence / 市场状态依赖性

We next examine whether spillover patterns differ across market states. We divide our sample into "high-volatility" and "low-volatility" regimes based on whether the VIX is above

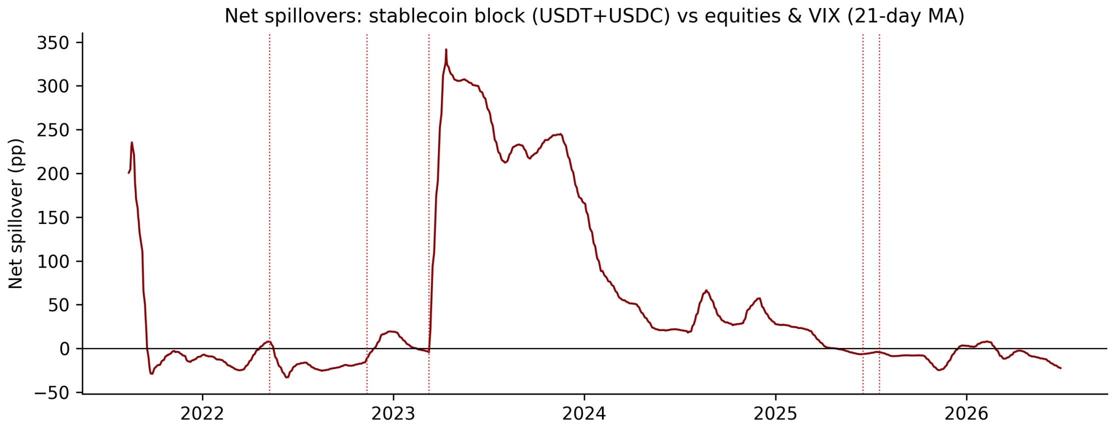

Figure 10: 稳定币板块相对股票与 VIX 的净溢出 / Net spillovers: stablecoin block (USDT+USDC) versus equities and VIX (21-day moving average). / 净溢出：稳定币板块（USDT+USDC）相对股票与 VIX（21 日移动平均）。

or below its sample median (17.9). Table 6 reports total and net spillovers across these regimes.

接下来我们考察溢出模式是否随市场状态而变化。我们根据 VIX 是否高于其样本中位数（17.9）将样本划分为“高波动”与“低波动”两种状态。表 6 报告了两种状态下的总溢出与净溢出。

Two findings stand out. First, the total spillover index is slightly lower in the high-VIX regime (44.8%) than in the low-VIX regime (49.6%). This does not contradict the crisis spikes documented above: regime averages pool prolonged calm stretches with short crisis bursts, whereas the TCI's crisis response is highly concentrated in the weeks immediately following discrete events. Second, the composition of net transmission shifts across regimes in an economically meaningful way. The VIX's net transmission nearly doubles in high-volatility states (from +30.2 to +57.4), while the stablecoins' net transmission weakens considerably (USDT from +15.8 to +1.8; USDC from +62.5 to +19.9). In other words, when market-wide uncertainty is high, the dominant driver of system-wide volatility is the equity uncertainty gauge itself, whereas in calmer states the stablecoin block retains a larger net transmitter role. Equity sectors remain net receivers in both regimes, with banks, payment, and crypto-concept stocks all receiving less volatility on net during high-VIX states.

有两点发现尤为突出。第一，高 VIX 状态下的总溢出指数（44.8%）略低于低 VIX 状态（49.6%）。这与上文记录的危机飙升并不矛盾：状态平均值将长时间的平静期与短暂的危机爆发混合在一起，而 TCI 对危机的反应高度集中在离散事件后的数周内。第二，净输出的构成在不同状态间以具有经济含义的方式转移。VIX 的净输出在高波动状态下几乎翻倍（从 +30.2 升至 +57.4），而稳定币的净输出明显减弱（USDT 从 +15.8 降至 +1.8；USDC 从 +62.5 降至 +19.9）。换言之，当全市场不确定性较高时，系统范围波动的主导驱动者是股票不确定性指标本身；而在较平静的状态下，稳定币板块保持着更大的净输出者角色。股票板块在两种状态下均为净接收者，且银行、支付和加密货币概念股在高 VIX 状态下的净接收规模都有所减小。

Table 6: 不同市场状态下的溢出 / Spillovers across market regimes

	Low VIX regime (VIX $<$ 17.9)	High VIX regime (VIX $\geq$ 17.9)
Total spillover index	49.59%	44.80%
Net spillovers:		
USDT	15.81	1.76
USDC	62.47	19.86
S&P 500	-36.25	-36.67
Banks	-18.04	-10.96
Exchanges	-3.21	-0.38
Payments	-21.46	-8.24
Crypto-concept	-29.54	-22.72
VIX	30.22	57.36

Notes: This table compares average spillover measures during low-volatility and high-volatility regimes, defined as VIX below and above its sample median of 17.9, respectively. All entries are in percentage points. 注：本表比较了低波动与高波动状态下的平均溢出测度，两种状态分别定义为 VIX 低于和高于其样本中位数 17.9。所有元素均以百分点表示。

7.3. Stablecoin-level heterogeneity / 稳定币层面的异质性

The two stablecoins in our system display distinct roles. USDC is the stronger net transmitter on average (NET = +40.9 versus +8.7 for USDT), a position almost entirely earned during and after the March 2023 depegging, when USDC volatility spilled over massively to the rest of the system. USDT, whose market capitalization is larger and whose price barely moved during the 2023 banking stress, transmits more evenly over time, and its short-frequency net transmission is the largest among all variables at the 1–5 day band (Table 4). This contrast is consistent with the two coins' different reserve compositions and user bases: USDC's reserves are concentrated in U.S. bank deposits and Treasuries, making it sensitive to U.S. banking stress, whereas USDT's offshore, more diversified reserve structure insulates it from U.S.-specific banking shocks.

我们系统中的两种稳定币表现出不同的角色。USDC 平均而言是更强的净输出者（NET = +40.9，USDT 为 +8.7），这一地位几乎完全是在 2023 年 3 月脱锚期间及之后形成的，当时 USDC 的波动向系统其余部分大规模溢出。USDT 的市值更大，且在 2023 年银行业压力期间其价格几乎未动，其输出在时间上分布更为均匀，并且在 1–5 天频段上是所有变量中最大的短期净输出者（见表 4）。这种差异与两种稳定币不同的储备构成与用户基础相一致：USDC 的储备集中于美国银行存款与国债，使其对美国银行业压力敏感；而 USDT 离岸化、更为分散的储备结构使其免受美国特定银行业冲击的影响。

8. Robustness checks / 稳健性检验

We conduct a battery of robustness checks to ensure our findings are not sensitive to specific modeling choices:

我们进行了一系列稳健性检验，以确保本文发现不依赖于特定的模型设定：

Alternative forecast horizons:. We re-estimate the spillover indices using forecast horizons of 5 and 20 days instead of our baseline 10 days. The total spillover index is 39.3% for $H = 5$ and 53.3% for $H = 20$, bracketing the baseline 47.2%, and the patterns of directional and net spillovers are unchanged.

替代预测期限：. 我们分别使用 5 天和 20 天的预测期限（基准为 10 天）重新估计溢出指数。总溢出指数在 $H = 5$ 时为 39.3%、在 $H = 20$ 时为 53.3%，将基准值 47.2% 包含其间，方向性溢出与净溢出的模式保持不变。

Alternative lag lengths:. We estimate TVP-VAR models with lag lengths 1 and 3. The total spillover index is 44.5% for $p = 1$ and 49.1% for $p = 3$, very close to the baseline estimate.

替代滞后阶数：. 我们估计了滞后阶数为 1 和 3 的 TVP-VAR 模型。总溢出指数在 $p = 1$ 时为 44.5%、在 $p = 3$ 时为 49.1%，与基准估计非常接近。

Alternative volatility estimator:. We replace GARCH-based conditional volatilities with range-based Garman-Klass volatilities computed from daily open, high, low, and close prices for the equity variables. The resulting total spillover index is 48.4%, essentially identical to the baseline.

替代波动率估计量：. 我们将基于 GARCH 的条件波动率替换为利用股票日度开盘价、最高价、最低价、收盘价计算的 Garman-Klass 极差波动率。所得总溢出指数为 48.4%，与基准结果基本一致。

Excluding crisis periods:. We re-estimate the static spillover table excluding the event windows around the Terra-LUNA collapse, the FTX bankruptcy, and the SVB/USDC depegging. The total spillover index falls only slightly to 47.1%, and the qualitative patterns—net transmission from the VIX and stablecoins, net reception by equities—are unchanged.

剔除危机时期：. 我们在剔除 Terra-LUNA 崩盘、FTX 破产和 SVB/USDC 脱锚前后的事件窗口后重新估计静态溢出表。总溢出指数仅小幅下降至 47.1%，且定性模式——VIX 与稳定币为净输出者、股票为净接收者——保持不变。

Placebo tests for tail risk:. We conduct placebo tests in which the stablecoin basket return series is randomly reshuffled and the CoVaR regressions are re-estimated. The average absolute placebo ΔCoVaR is 0.35%, far below the estimated magnitudes for the crypto-concept (1.26%) and exchange (1.06%) portfolios and roughly half the average across all sectors (0.61%), confirming that our tail risk results are not spurious.

尾部风险的安慰剂检验：. 我们进行了安慰剂检验，将稳定币篮子收益序列随机打乱后重新估计 CoVaR 回归。安慰剂 ΔCoVaR 的平均绝对值为 0.35%，远低于加密货币概念组合（1.26%）和交易所组合（1.06%）的估计值，约为所有板块平均值（0.61%）的一半，证实本文的尾部风险结果并非虚假相关。

9. Conclusion and policy implications / 结论与政策含义

This paper provides a comprehensive analysis of volatility spillovers and extreme risk contagion between stablecoin markets and U.S. stock markets across disaggregated sectors. Using TVP-VAR connectedness measures, frequency-domain decomposition, and CoVaR/MES tail risk metrics on a daily sample from April 2021 to June 2026, we document several novel findings.

本文全面分析了稳定币市场与美国股票市场在细分板块层面的波动溢出与极端风险传染。基于 2021 年 4 月至 2026 年 6 月的日度样本，利用 TVP-VAR 关联性测度、频域分解以及 CoVaR/MES 尾部风险指标，我们记录了若干新发现。

First, stablecoin markets are materially connected to equity markets, with an average total spillover index of 47.2%. Second, connectedness is highly time-varying: the largest spike in the sample—a jump to 87.5%—occurred during the March 2023 SVB failure and USDC depegging, dwarfing the responses to the crypto-native Terra-LUNA and FTX crises, while the GENIUS Act legislative events left connectedness broadly unchanged. Third, spillovers are dominated by the long-term (21+ days) frequency component, which accounts for nearly half of total connectedness, indicating persistent regime-level linkages rather than transient high-frequency transmission. Fourth, tail risk exposure is sharply heterogeneous across sectors: crypto-concept and exchange stocks exhibit large and statistically significant ΔCoVaR and MES, payment stocks and the broad market show moderate exposure, and bank stocks show none. Finally, the stablecoin block is a net volatility transmitter to equities throughout the sample, and its transmitter role strengthens during crisis windows.

第一，稳定币市场与股票市场存在实质性关联，平均总溢出指数为 47.2%。第二，关联性高度时变：样本中最大的一次飙升——跳升至 87.5%——发生在 2023 年 3 月 SVB 倒闭与 USDC 脱锚期间，远超对 Terra-LUNA 和 FTX 等加密原生危机的反应，而《GENIUS 法案》立法事件则基本没有改变关联水平。第三，溢出由长期（21 天以上）频率成分主导，其占总关联性的近一半，表明存在持久的状态层面联系而非短暂的高频传导。第四，尾部风险暴露在板块间高度异质：加密货币概念股与交易所股票表现出巨大且统计显著的 ΔCoVaR 与 MES，支付股与大盘的暴露居中，银行股则无显著暴露。最后，稳定币板块在整个样本期内都是对股市的波动净输出者，且其输出者角色在危机窗口进一步增强。

Our findings have several important policy implications. First, the largest cross-market spillover episode in our sample originated not in the crypto ecosystem but in the traditional banking system: the SVB failure propagated to equity markets through the USDC peg. This directly supports regulatory requirements—as contemplated in the GENIUS Act—for high-quality, transparent, and segregated stablecoin reserve assets, since the perceived safety of reserves is the key margin along which banking stress can be transmitted into digital asset markets and back into equities. Second, the concentration of tail risk exposure in crypto-concept and exchange stocks suggests that macroprudential oversight should pay particular attention to listed firms with significant crypto-related business lines, as these constitute the primary transmission channel of stablecoin stress into public equity markets. Third, the dominance of low-frequency spillovers implies that stablecoin-equity linkages are structural rather than episodic, and are likely to deepen as regulated stablecoin issuance by banks and payment companies expands under the new legislative framework.

本文发现具有若干重要政策含义。第一，样本中最大的一次跨市场溢出事件并非源自加密生态系统，而是源自传统银行体系：SVB 倒闭通过 USDC 锚定传导至股票市场。这

直接支持《GENIUS 法案》所设想的对高质量、透明、隔离的稳定币储备资产的监管要求，因为储备的感知安全性是银行业压力传入数字资产市场并回传至股票市场的关键边际渠道。第二，尾部风险暴露集中于加密货币概念股与交易所股票，表明宏观审慎监管应特别关注拥有大量加密相关业务的上市公司，因为它们构成了稳定币压力传入公开股票市场的主要传导渠道。第三，低频溢出的主导地位意味着稳定币与股票市场的联系是结构性的而非偶发性的，并且随着新立法框架下银行和支付公司受监管稳定币发行业务的扩张，这种联系可能进一步加深。

For portfolio managers, our results indicate that crypto-concept and exchange stocks offer little diversification benefit with respect to stablecoin market risks, particularly in the left tail, while traditional bank stocks are effectively insulated from stablecoin tail shocks during our sample period. The strong time variation in connectedness—and its concentration around peg-threatening events—suggests that dynamic, event-aware hedging strategies are required to manage stablecoin-related risks in equity portfolios.

对于投资组合管理者而言，本文结果表明加密货币概念股与交易所股票在稳定币市场风险方面几乎不提供分散化收益，尤其是在左尾；而传统银行股在本文样本期内实际上与稳定币尾部冲击相隔离。关联性的强时变性——及其集中于威胁锚定的事件周围——表明在股票组合中管理稳定币相关风险需要动态的、事件感知的对冲策略。

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